



Submersible Borehole Pump

UPA S

Size: UPA S 250

Nominal Pump Diameter: 10 Inches

Type Series Booklet



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Type Series Booklet UPA S

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Water Supply

Submersible Borehole Pumps

UPA S 250



Main applications

- Spray irrigation systems
- General irrigation systems
- Pressure boosting
- Groundwater management
- Water supply systems
- Offshore processes

Fluids handled

- Drinking water
- River water, lake water and groundwater
- Maximum sand content in fluid handled depending on particle size and duration, sand content up to 250 g/m³¹⁾

Operating data

Table 1: Operating properties

Characteristic		Value		
		50 Hz	60 Hz	100 Hz
Flow rate	Q [m ³ /h]	≤ 383	≤ 462	≤ 396
	Q [l/s]	≤ 106	≤ 128	≤ 110
Head	H [m]	≤ 373	≤ 543	≤ 399
Fluid temperature	T [°C]	≤ 50	≤ 50	≤ 50
Speed	n [rpm]	≤ 2900	≤ 3500	≤ 3000
Well diameter	D [mm]	≥ 250	≥ 250	≥ 250
	D ["]	≥ 10	≥ 10	≥ 10

Design details

Design

- Centrifugal pump
- Single-stage or multistage
- Single-suction
- Ring-section design
- Rigid connection between pump and motor

Type of installation

- Vertical installation
- Horizontal installation
- Angled installation

Drive

Asynchronous motor:

- With squirrel cage motor for submerged use
- Standard connection to NEMA
- Frequency 50 Hz/60 Hz
- IP68 enclosure
- DOL starting or star-delta starting
- Frequency of starts ≤ 15 starts per hour
- Winding J1 (PPC/HDPE) or, for higher temperatures, J2 (radiation-crosslinked PE2/PA)

Synchronous motor:

- Interior (buried) permanent magnet synchronous motor (IPMSM)
- IP68 enclosure
- Standard connection to NEMA
- Frequency 100 Hz
- Winding J2 (radiation-crosslinked PE2/PA)
- Frequency of starts ≤ 15 starts per hour

Electrical connection

- Delivered ex-factory with 1 or 2 motor leads (including earth conductor and internal earthing)
- Extension cable connected with water-tight cable connector
- Motor lead and extension cable suitable for drinking water use

Impeller type

- Mixed flow design

Bearings

- Radial plain bearings
- Pump bearings lubricated by fluid handled; motor bearings lubricated by water fill
- Axial thrust is balanced by a tilting-pad thrust bearing in the motor (lower end)

¹ For 50 g/m³ and above, the more wear-resistant version with silicon carbide bearings is recommended.

Connections

- Pump screw-ended or flanged
- Threaded end to DIN ISO 228, Part 1 (up to 6 inch)
- NPT threaded end to ASME B1.20.1-2013
- Flanged end to DIN EN 1092
- With lift check valve or connection nozzle

Designation

Table 2: Designation example

Position																																
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	
U	P	A	S		2	5	0	-	2	4	0	/	0	3	C	C	+	U	M	A	S	2	0	0	-	7	5	/	4	2	C	
Pump																	+	Motor														
Pump set																																

Table 3: Designation key

Position	Code	Description
1-5	Type series	
	UPA S	Precision (investment) cast variant
6-8	Nominal pump diameter [mm]	
	250	
9-12	Flow rate at best efficiency point [m³/h]	
	-140	140
	-190	190
	-240	240
13	/	
14-15	Number of stages	
	01, 02, 03, 04, 05, 06, 07, 08, 09	
16	Pump material	
	D	Duplex 1.4517
	C	Stainless steel AISI 316 SS
17	Impeller material	
	D	Duplex 1.4517
	C	Stainless steel AISI 316 SS
18	+	With motor
	-	Without motor
19-22	Motor version	
	UMA	Asynchronous motor
	UMA S	Permanent magnet synchronous motor
23-25	Minimum motor diameter [mm]	
	150, 200, 250, 300	
26	-	
27-28	Motor rating PN [kW]	
	18	18,5
	...	...
	250	250
29	/	
30	Number of motor poles	
	2	2 poles
	4	4 poles
31	Winding insulation	
	1	J1 (PPC/HDPE)
	2	J2 (radiation-crosslinked PE2/PA)
32	Motor material	
	E	Stainless steel AISI 304 SS
	D	1.4539 / AISI 904L
	C	Stainless steel AISI 316 SS
	G	Grey cast iron

Materials

Table 4: Pump material selection, UPA S 250

Component	Material variant	
	C	D
Impeller	CrNiMo steel (1.4408)	CrNiMo steel (1.4517)
Suction casing	CrNiMo steel (1.4408)	CrNiMo steel (1.4517)
Screws, bolts and nuts	CrNiMo steel (A4)	CrNiMo steel (1.4462)
Stage casing	CrNiMo steel (1.4408)	CrNiMo steel (1.4517)
Shaft	CrNiMo steel (1.4462)	CrNiMo steel (1.4462)

Table 5: Motor material selection, UMA 150, UMA-S 150, UMA 200, UMA-S 200, UMA 250, UMA-S 250, UMA 300

Component	Size	Material variant			
		G	E	C	D
Housing	UMA 150, UMA-S 150	-	CrNi steel (1.4301)	CrNiMo steel (1.4571)	CrNiMo steel (1.4539)
	UMA 200, UMA-S 200	Grey cast iron (EN-GJL-200)	-	CrNiMo steel (1.4408)	CrNiMo steel (1.4539)
	UMA 250, UMA-S 250	Grey cast iron (EN-GJL-200)	-	CrNiMo steel (1.4408)	CrNiMo steel (1.4539)
	UMA 300	Grey cast iron (EN-GJL-250)	-	-	CrNiMo steel (1.4517)
Screws, bolts and nuts	UMA 150, UMA-S 150	-	CrNiMo steel (A4-70)	CrNiMo steel (A4-70)	CrNiMo steel (1.4539)
	UMA 200, UMA-S 200	CrNiMo steel (A4-70)	-	CrNiMo steel (A4-70)	CrNiMo steel (1.4539)
	UMA 250, UMA-S 250	CrNiMo steel (A4-70)	-	CrNiMo steel (A4-70)	CrNiMo steel (1.4539)
	UMA 300	CrNiMo steel (1.4571)	-	-	CrNiMo steel (1.4462)
Shaft	UMA 150	-	CrNi steel (1.4021)	CrNiMo steel (1.4462)	CrNiMo steel (1.4462)
	UMA-S 150	-	CrNiMo steel (1.4462)	CrNiMo steel (1.4462)	CrNiMo steel (1.4462)
	UMA 200, UMA-S 200	CrNiMo steel (1.4460)	-	CrNiMo steel (1.4460)	CrNiMo steel (1.4462)
	UMA 250, UMA-S 250	CrNiMo steel (1.4460)	-	CrNiMo steel (1.4460)	CrNiMo steel (1.4462)
	UMA 300	CrNiMo steel (1.4462-C45+N)	-	-	CrNiMo steel (1.4462-C45+N)

Product benefits

- High efficiencies through optimised, precision (investment) cast hydraulic design
- High operating reliability as all components are made of high-quality stainless steel.
- Overall efficiency can be increased further by combining the pump with an UMA S synchronous motor with ratings of up to 250 kW
- Reliable operation by anti-jam lift check valve
- Easy to install with service-friendly hydraulic system design
- Solids-induced wear can be minimised by metal wear rings and bearings with enhanced wear resistance (optional).
- Patented intake chamber provides optimum flow conditions at low heads.

Certifications

Table 6: Overview

Label	Effective in:	Comment
	All countries	Certified quality management to ISO 9001
	France	Approved in accordance with the French drinking water regulation

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see <https://www.ksb.com/en-global/company/corporate-responsibility/reach>.

Overview of submersible motors

The following submersible motors are available:

Table 7: Selection table

Well diameter	Motor size	Voltage range	Rated power
6 inches	UMA 150	Up to 1 kV	5.5 to 37 kW
6 inches	UMA-S 150	Up to 1 kV	4.0 to 37 kW
8 inches	UMA 200	Up to 1 kV	37 to 90 kW
8 inches	UMA-S 200	Up to 1 kV	45 to 130 kW

Well diameter	Motor size	Voltage range	Rated power
10 inches	UMA 250	Up to 1 kV	85 to 190 kW
10 inches	UMA-S 250	Up to 1 kV	150 to 250 kW
12 inches	UMA 300	Up to 1 kV	250 to 400 kW

- Further motor data for the UMA asynchronous motor see type series booklet No. 3455.51
- Further motor data for the UMA-S synchronous motor see type series booklet No. 3455.52

Technical data, 50 Hz

Information on characteristic curves

The characteristic curves shown are intended to allow pre-selection. Refer to the quotation for the precise selection data.

Tolerance to ISO 9906 Cl. 2B

The characteristic curves shown are valid for:

- Water at 20 °C
- Density 998 kg/m³
- Viscosity 1.00 mm²/s

Selection chart UPA S 250, 50 Hz, n = 2900 rpm

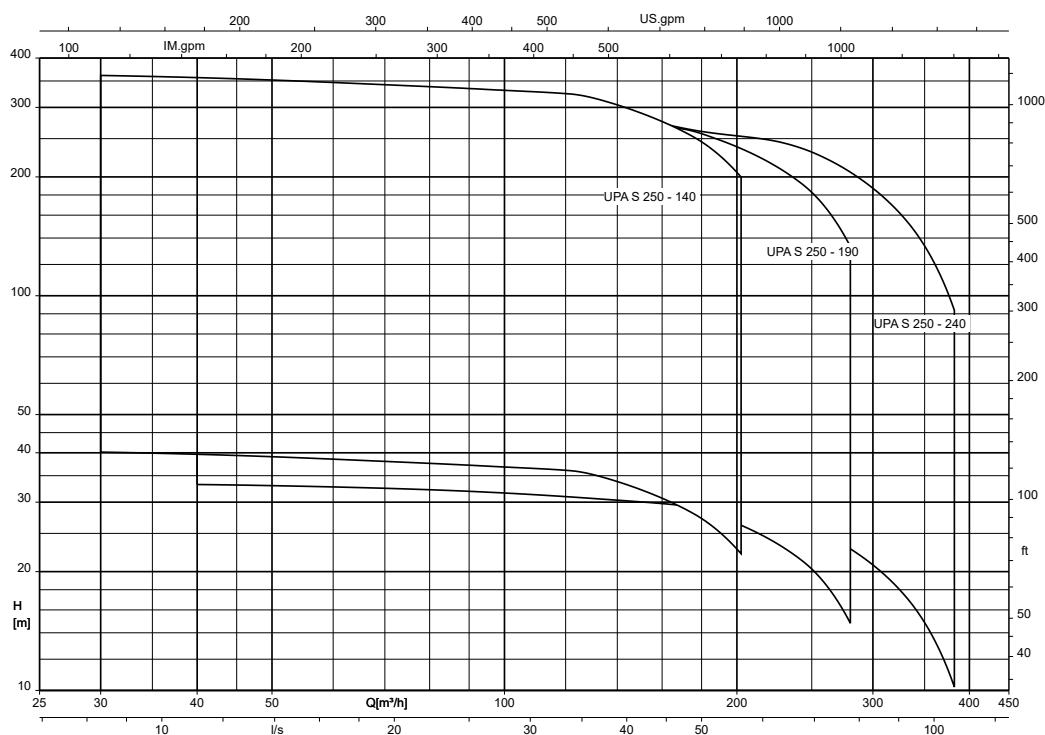
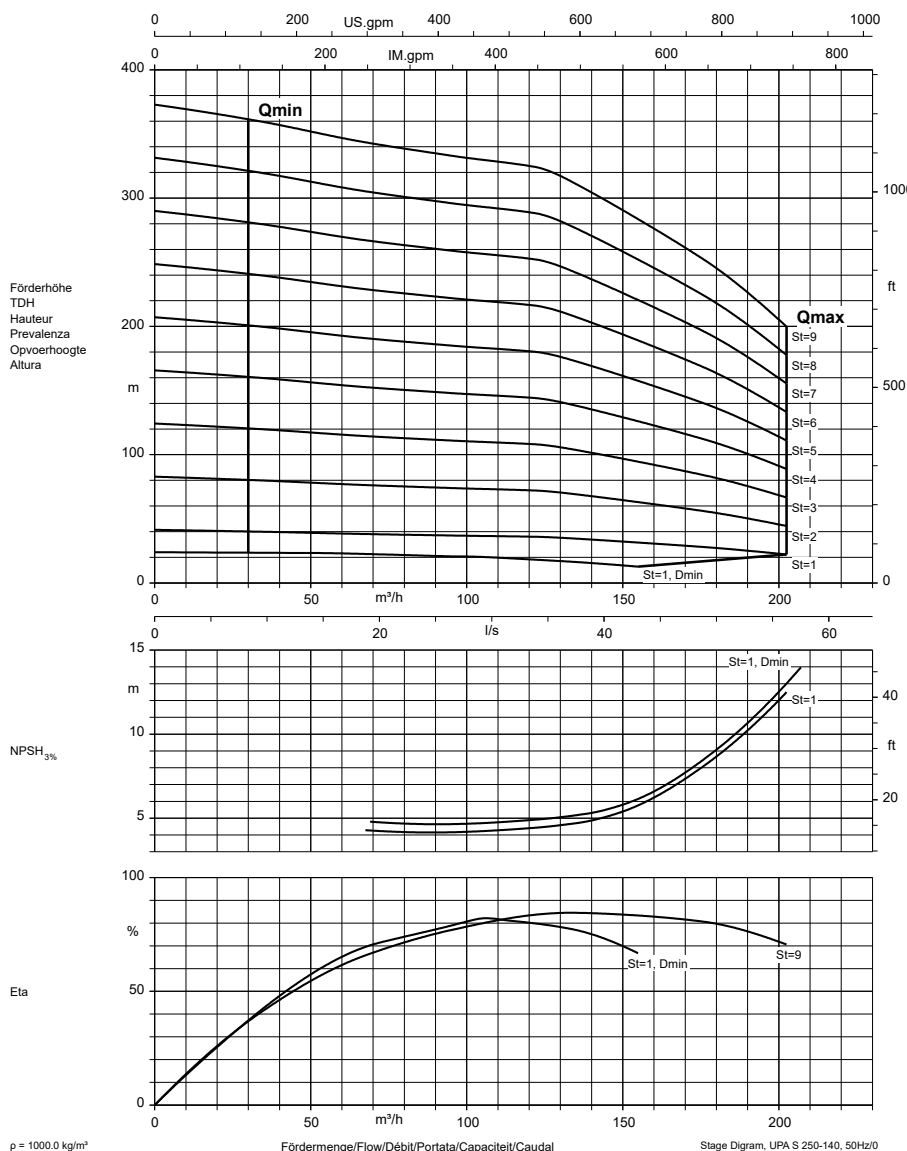


Fig. 1: Selection chart UPA S 250, 50 Hz

UPA S 250 - 140 / ..., number of stages 1 - 9, 50 Hz, n = 2900 rpm



Connection types

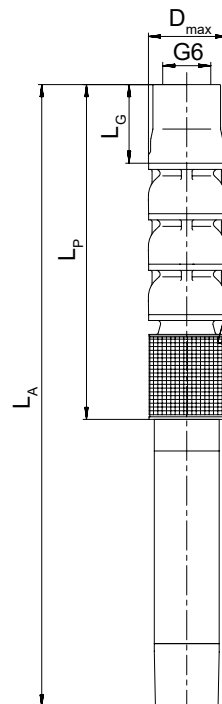
Standard pump end = G6

Alternative pump ends:

- G8
- DN 150, DN 200
- NPT6-8, NPT8-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 35)



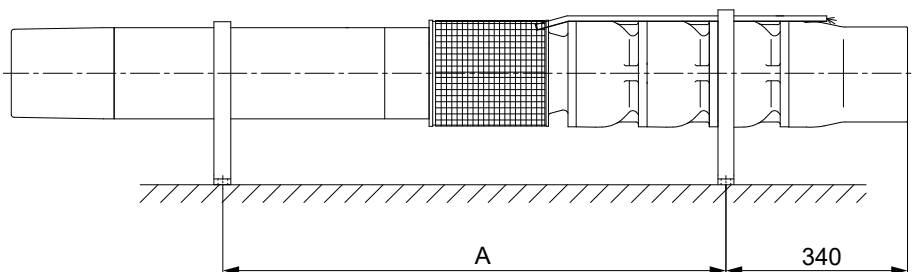
Operating range

$Q_{min} = 30 \text{ m}^3/\text{h}$

Q_{max} = end of stage curve

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 4)



Dimensions of UPA S 250 - 140 / ... [mm]

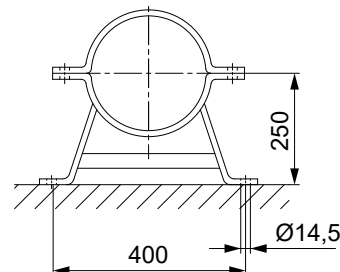


Table 8: Dimensions, weights and installation type depending on the motor, 50 Hz / 400 V 3~

UPA S 250 - 140 / ...	A	L _P ²⁾	L _A ²⁾	L _G ²⁾	D _{max} ²⁾		Total weight	Type of installation	
					DOL	Y - Δ			
					[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]	Vertical	Horizontal
1 + UMA 150 18/21	741	631	1530	210	243	247	125	X	X
2 + UMA 150 37/22	-	787	2061	210	247	247	181	X	-
2 + UMA 200 37/21	1017	787	1927	210	247	247	221	X	X
3 + UMA 200 55/21	1273	943	2283	210	247	247	276	X	X
4 + UMA 200 75/21	1539	1099	2659	210	246	246	334	X	X
5 + UMA 200 90/21	-	1255	2995	210	246	246	385	X	-
4 + UMA 250 85/21	1509	1139	2558	210	249	249	405	X	X
5 + UMA 250 110/21	-	1295	2824	210	249	253	461	X	-
6 + UMA 250 110/21	-	1451	2980	210	249	253	480	X	-
7 + UMA 250 132/21	-	1607	3266	210	253	253	543	X	-
8 + UMA 250 160/21	-	1763	3532	210	253	270	599	X	-
9 + UMA 250 190/21	-	1919	3838	210	270	270	669	X	-

Table 9: Technical data, 50 Hz / 400 V 3~, n = 2900 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 250 - 140 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m³/h	Maximum rated power	Rated power	Rated current	Efficiency	Number × cross-section of cores ³⁾	
	H ₀	P ₂	P _N	I _N	η _M	DOL	Y - Δ
	[m]	[kW]	[kW]	[A]	[%]	[mm²]	[mm²]
1 + UMA 150E 18/21	41	17	19	42	81,5	F4 × 4	F3/F4 × 2,5
2 + UMA 150E 37/22	83	35	37	82	83,5	F3/F4 × 4 II	F3/F4 × 4
2 + UMA 200D 37/21	83	35	37	75	85,0	F4 × 10	F3/F4 × 6
3 + UMA 200D 55/21	124	52	55	110	87,0	F4 × 16	F3/F4 × 6
4 + UMA 200D 75/21	166	69	75	151	87,5	F3/F4 × 10 ⁴⁾	F3/F4 × 10
5 + UMA 200D 90/21	207	87	90	180	88,0	F3/F4 × 16 ⁴⁾	F3/F4 × 16
4 + UMA 250D 85/21	166	69	85	164	87,5	F3/F4 × 10 ⁴⁾	F3/F4 × 16
5 + UMA 250D 110/21	207	87	110	216	88,5	F3/F4 × 16 ⁴⁾	F3/F4 × 25
6 + UMA 250D 110/21	249	104	110	216	88,5	F3/F4 × 16 ⁴⁾	F3/F4 × 25
7 + UMA 250D 132/21	290	121	132	249	89,0	F3/F4 × 25 ⁴⁾	F3/F4 × 25
8 + UMA 250D 160/21	331	139	160	301	89,0	F3/F4 × 25 II	F3/R4 × 35
9 + UMA 250D 190/21	373	156	190	365	89,5	F3/R4 × 35 II	F3/R4 × 35

Table 10: Explanation of abbreviations

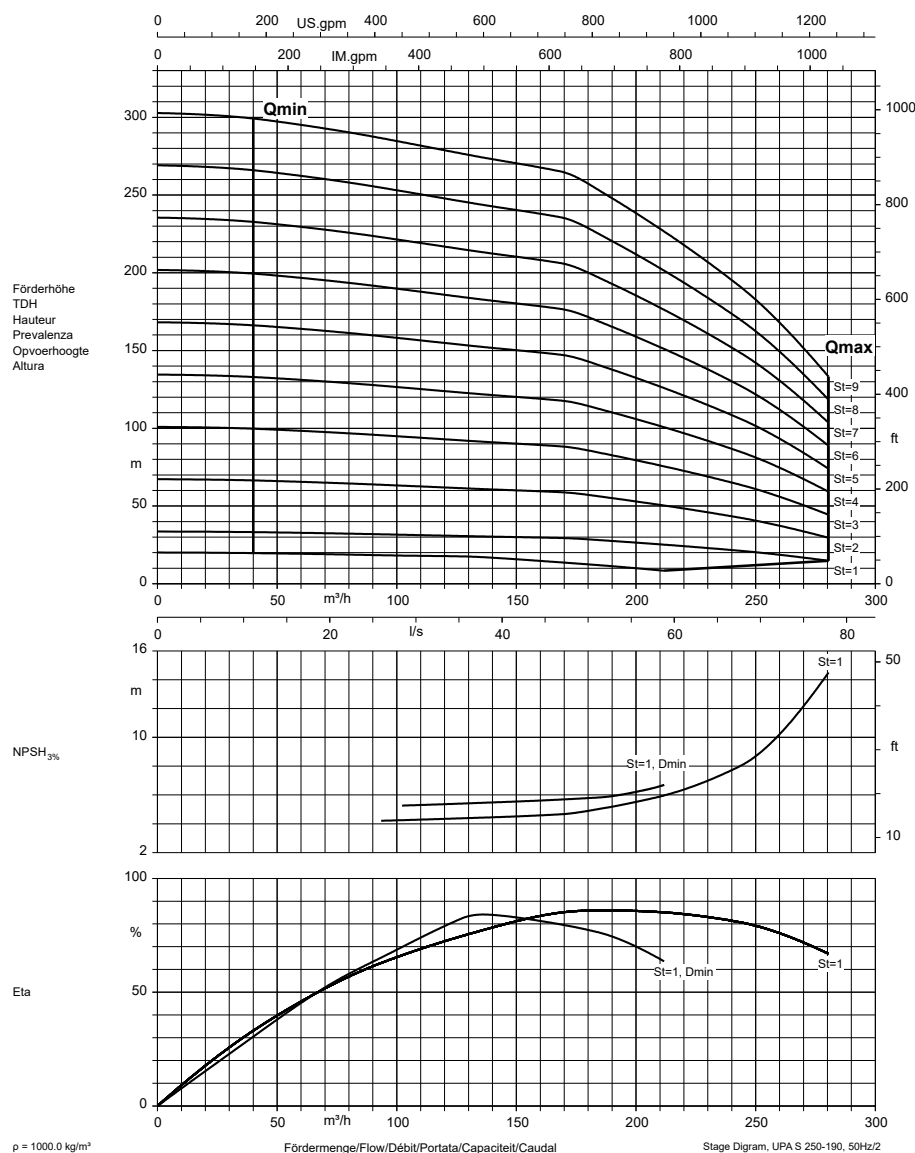
Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

²⁾ Dimension indicated refers to version with standard pump end G6.

³⁾ Designed for submerged use, 400 V, ≤ +30 °C

⁴⁾ Delta configuration in cable connector with extension cable or in control cabinet

UPA S 250 - 190 / ..., number of stages 1 - 9, 50 Hz, n = 2900 rpm

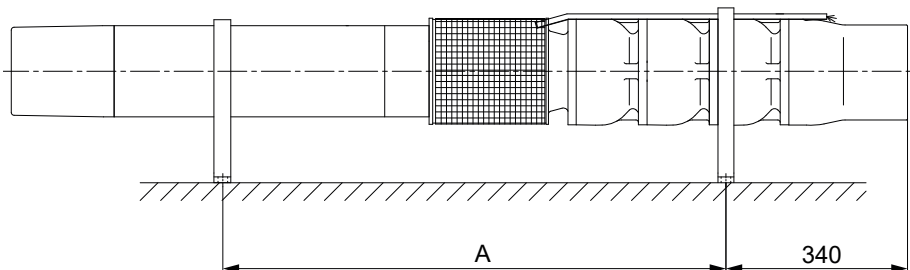


Operating range

$Q_{min} = 40 \text{ m}^3/\text{h}$
 $Q_{max} = \text{end of stage curve}$

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 5)



Dimensions of UPA S 250 - 190 / ... [mm]

Connection types

Standard pump end = G6

Alternative pump ends:

- G8
- DN 150, DN 200
- NPT6-8, NPT8-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 36)

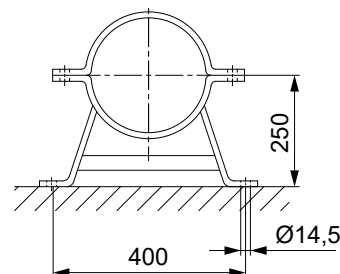
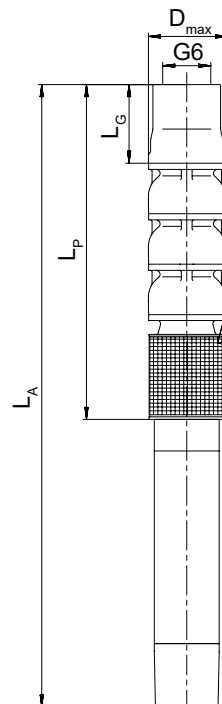


Table 11: Dimensions, weights and installation type depending on the motor, 50 Hz / 400 V 3~

UPA S 250 - 190 / ...	A	L _P ⁵⁾	L _A ⁵⁾	L _G ⁵⁾	D _{max} ⁵⁾		Total weight	Type of installation	
					DOL	Y - Δ		Vertical	Horizontal
					[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]		
1 + UMA 150 18/21	753	643	1542	210	243	247	125	X	X
2 + UMA 150 37/22	-	811	2085	210	247	247	182	X	-
2 + UMA 200 37/21	1041	811	1951	210	247	247	222	X	X
3 + UMA 200 55/21	1309	979	2319	210	247	247	278	X	X
4 + UMA 200 75/21	1587	1147	2707	210	246	246	337	X	X
5 + UMA 200 90/21	-	1315	3055	210	246	246	388	X	-
4 + UMA 250 85/21	1557	1187	2606	210	249	249	407	X	X
5 + UMA 250 110/21	-	1355	2884	210	249	253	464	X	-
6 + UMA 250 110/21	-	1523	3052	210	249	253	484	X	-
7 + UMA 250 132/21	-	1691	3350	210	253	253	547	X	-
8 + UMA 250 160/21	-	1859	3628	210	253	270	605	X	-
9 + UMA 250 190/21	-	2027	3946	210	270	270	675	X	-

Table 12: Technical data, 50 Hz / 400 V 3~, n = 2900 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 250 - 190 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m³/h	Maximum rated power	Rated power	Rated current	Efficiency	Number × cross-section of cores ⁶⁾	
	H ₀	P ₂	P _N	I _N	η _M	DOL	Y - Δ
	[m]	[kW]	[kW]	[A]	[%]	[mm²]	[mm²]
1 + UMA 150E 18/21	34	17	19	42	81,5	F4 x 4	F3/F4 x 2.5
2 + UMA 150E 37/22	67	35	37	82	83,5	F3/F4 x 4 II	F3/F4 x 4
2 + UMA 200D 37/21	67	35	37	75	85,0	F4 x 10	F3/F4 x 6
3 + UMA 200D 55/21	101	52	55	110	87,0	F4 x 16	F3/F4 x 6
4 + UMA 200D 75/21	135	70	75	151	87,5	F3/F4 x 10 ⁷⁾	F3/F4 x 10
5 + UMA 200D 90/21	168	87	90	180	88,0	F3/F4 x 16 ⁷⁾	F3/F4 x 16
4 + UMA 250D 85/21	135	70	85	164	87,5	F3/F4 x 10 ⁷⁾	F3/F4 x 16
5 + UMA 250D 110/21	168	87	110	216	88,5	F3/F4 x 16 ⁷⁾	F3/F4 x 25
6 + UMA 250D 110/21	202	105	110	216	88,5	F3/F4 x 16 ⁷⁾	F3/F4 x 25
7 + UMA 250D 132/21	235	122	132	249	89,0	F3/F4 x 25 ⁷⁾	F3/F4 x 25
8 + UMA 250D 160/21	269	140	160	301	89,0	F3/F4 x 25 II	F3/R4 x 35
9 + UMA 250D 190/21	303	157	190	365	89,5	F3/R4 x 35 II	F3/R4 x 35

Table 13: Explanation of abbreviations

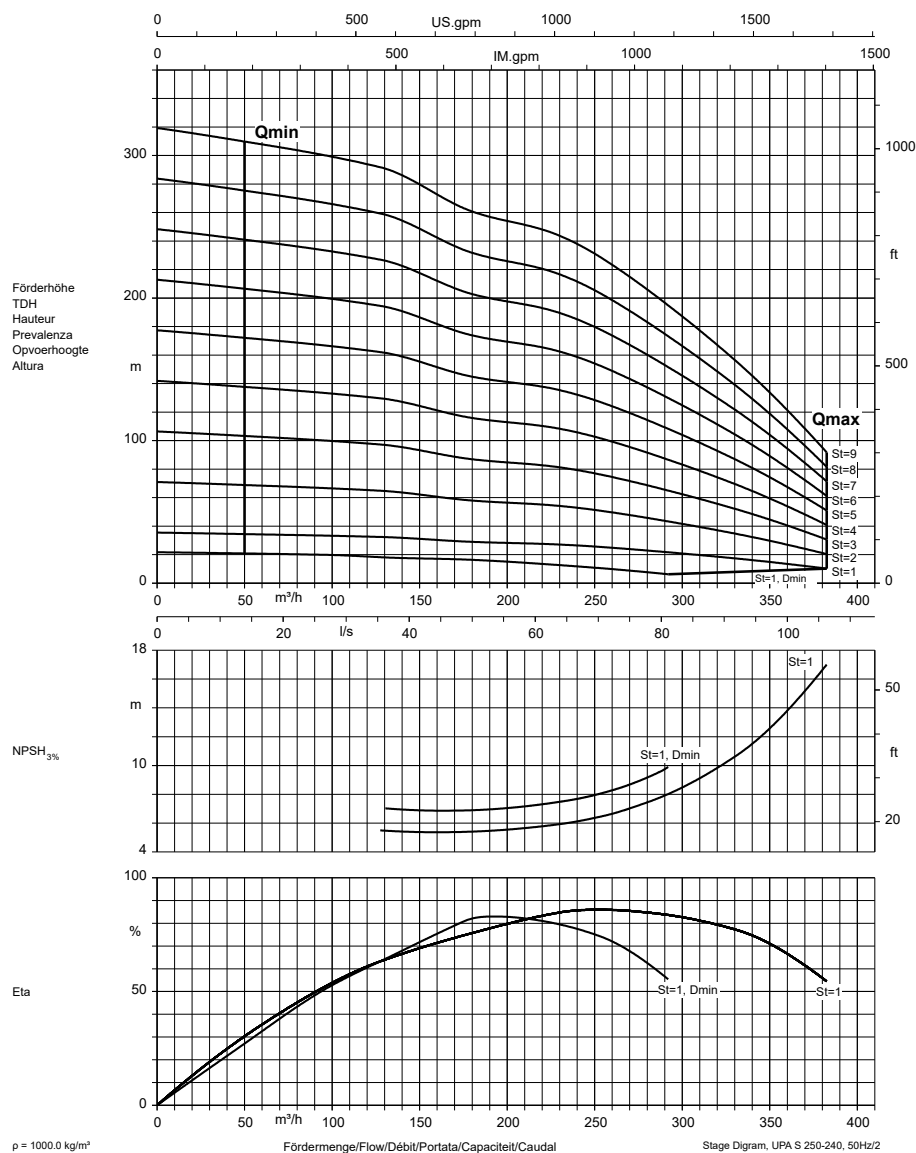
Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

5 Dimension indicated refers to version with standard pump end G6.

6 Designed for submerged use, 400 V, ≤ +30 °C

7 Delta configuration in cable connector with extension cable or in control cabinet

UPA S 250 - 240 / ..., number of stages 1 - 9, 50 Hz, n = 2900 rpm



Connection types

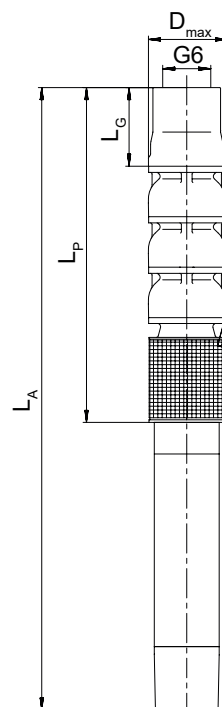
Standard pump end = G6

Alternative pump ends:

- G8
- DN 150, DN 200
- NPT6-8, NPT8-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 37)



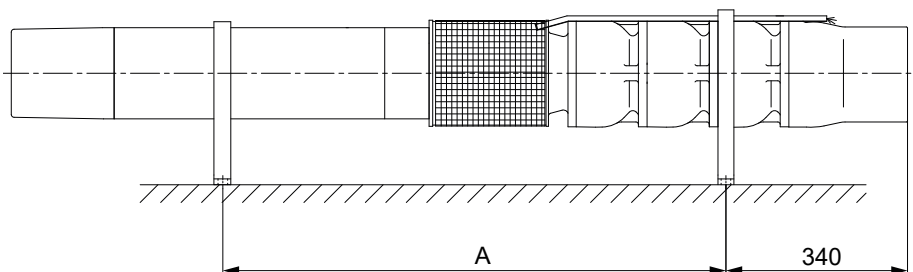
Operating range

$Q_{min} = 50 \text{ m}^3/\text{h}$

Q_{max} = end of stage curve

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 6)



Dimensions of UPA S 250 - 240 / ... [mm]

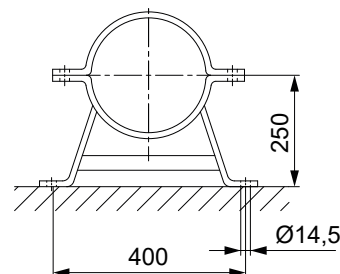


Table 14: Dimensions, weights and installation type depending on the motor, 50 Hz / 400 V 3~

UPA S 250 - 240 / ...	A	L _P ⁸⁾	L _A ⁸⁾	L _G ⁸⁾	D _{max} ⁸⁾		Total weight	Type of installation	
					DOL	Y - Δ		Vertical	Horizontal
					[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]		
1 + UMA 150 22/21	803	648	1637	210	242	246	133	✗	✗
2 + UMA 200 45/21	1096	821	2051	210	246	246	236	✗	✗
3 + UMA 200 65/21	1389	994	2464	210	246	246	298	✗	✗
4 + UMA 200 90/21	-	1167	2907	210	245	245	365	✗	-
4 + UMA 250 110/21	1632	1207	2736	210	248	253	440	✗	✗
5 + UMA 250 110/21	-	1380	2909	210	248	253	459	✗	-
6 + UMA 250 132/21	-	1553	3212	210	253	253	522	✗	-
7 + UMA 250 160/21	-	1726	3495	210	253	270	578	✗	-
8 + UMA 250 190/21	-	1899	3818	210	270	270	648	✗	-
9 + UMA 300 250/22	-	2072	4145	210	283	283	794	✗	-

Table 15: Technical data, 50 Hz / 400 V 3~, n = 2900 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 250 - 240 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m³/h	Maximum rated power	Rated power	Rated current	Efficiency	Number × cross-section of cores ⁹⁾	
						DOL	Y - Δ
						[mm²]	[mm²]
1 + UMA 150E 22/21	35	21	22	49	83,0	F4 × 4	F3/F4 × 2.5
2 + UMA 200D 45/21	71	41	45	90	86,0	F4 × 10	F3/F4 × 6
3 + UMA 200 65/21	106	62	65	129	87,5	F3/F4 × 10 ¹⁰⁾	F3/F4 × 10
4 + UMA 200D 90/21	142	83	90	180	88,0	F3/F4 × 16 ¹⁰⁾	F3/F4 × 16
4 + UMA 250D 110/21	142	83	110	216	88,5	F3/F4 × 16 ¹⁰⁾	F3/F4 × 25
5 + UMA 250D 110/21	177	104	110	216	88,5	F3/F4 × 16 ¹⁰⁾	F3/F4 × 25
6 + UMA 250D 132/21	213	124	132	249	89,0	F3/F4 × 25 ¹⁰⁾	F3/F4 × 25
7 + UMA 250D 160/21	248	145	160	301	89,0	F3/F4 × 25 II	F3/R4 × 35
8 + UMA 250D 190/21	284	166	190	365	89,5	F3/R4 × 35 II	F3/R4 × 35
9 + UMA 300D 250/22	319	186	250	466	88,5	6×R1×95 II + R1×95	6×R1×95 II + R1×95

Table 16: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
R1	1 × 1-core, cable cross-section: round
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

8 Dimension indicated refers to version with standard pump end G6.

9 Designed for submerged use, 400 V, ≤ +30 °C

10 Delta configuration in cable connector with extension cable or in control cabinet

Technical data, 60 Hz

Tolerance to ISO 9906 Cl. 2B

Information on characteristic curves

The characteristic curves shown are intended to allow pre-selection. Refer to the quotation for the precise selection data.

The characteristic curves shown are valid for:

- Water at 20 °C
- Density 998 kg/m³
- Viscosity 1.00 mm²/s

Selection chart UPA S 250, 60 Hz, n = 3500 rpm

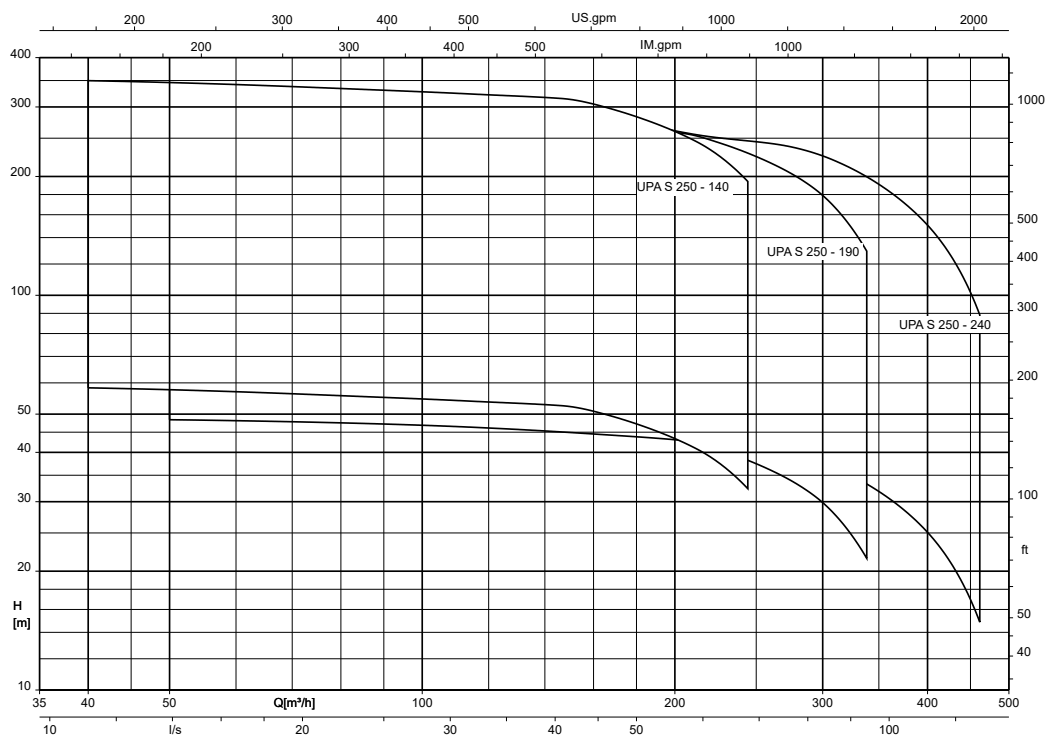
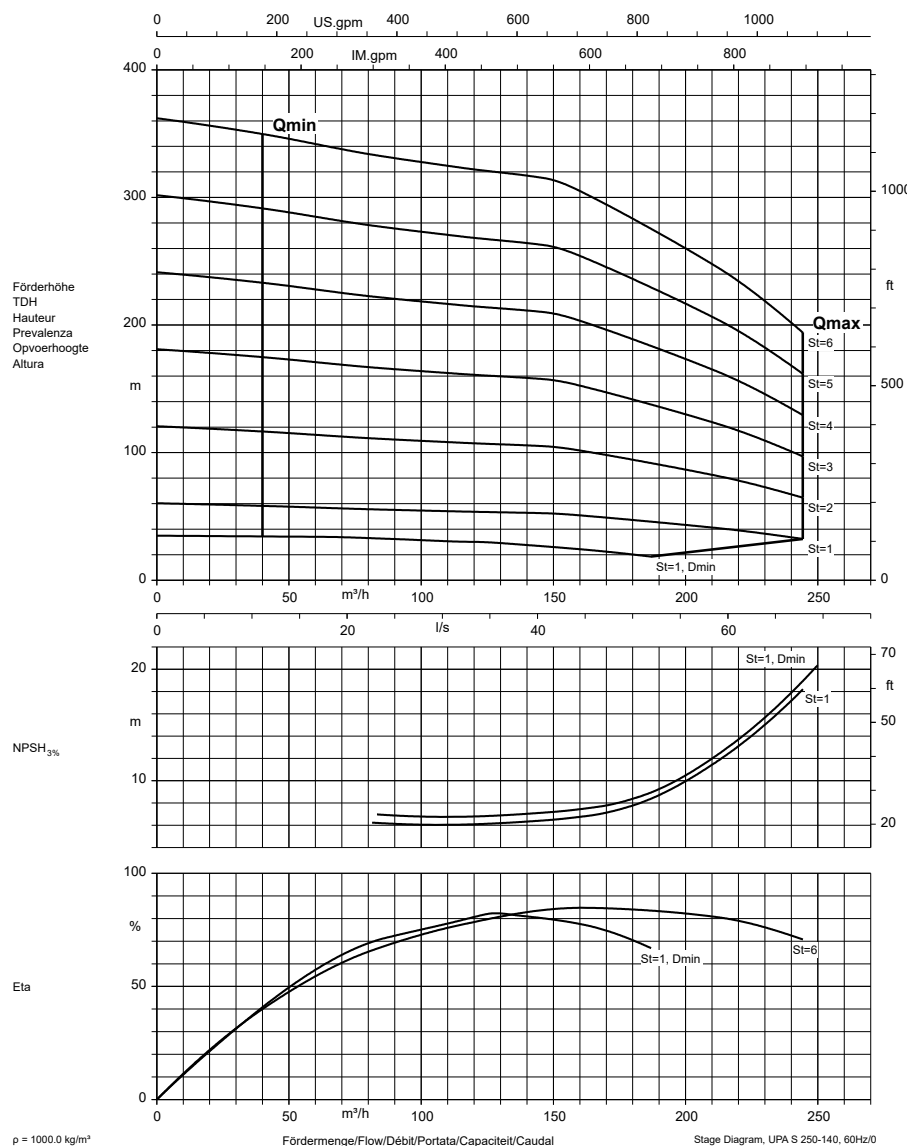


Fig. 2: Selection chart UPA S 250, 60 Hz

UPA S 250 - 140 / ..., number of stages 1 - 6, 60 Hz, n = 3500 rpm



Connection types

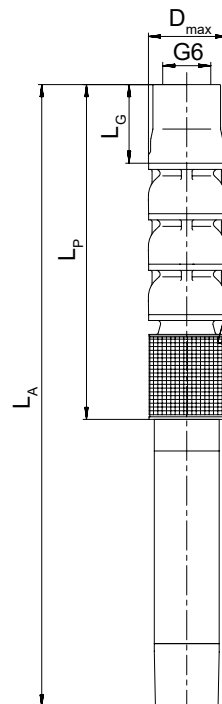
Standard pump end = G6

Alternative pump ends:

- G8
- DN 150, DN 200
- NPT6-8, NPT8-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 35)



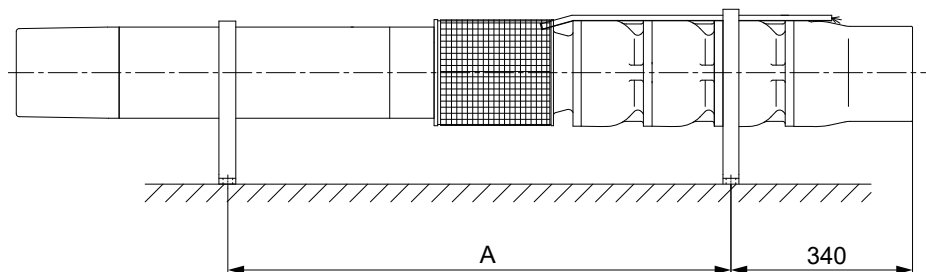
Operating range

$Q_{min} = 40 \text{ m}^3/\text{h}$

Q_{max} = end of stage curve

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 4)



Dimensions of UPA S 250 - 140 / ... [mm]

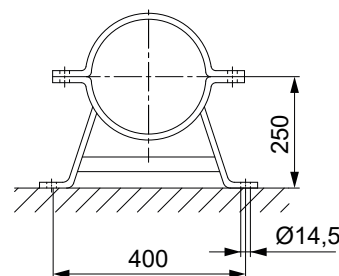


Table 17: Dimensions, weights and installation type depending on the motor, 60 Hz / 460 V 3~

UPA S 250 - 140 / ...	A	L _p ¹¹⁾	L _A ¹¹⁾	L _G ¹¹⁾	D _{max} ¹¹⁾		Total weight	Type of installation	
					DOL	Y - Δ			
					[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]	Vertical	Horizontal
1 + UMA 150 30/21	888	631	1825	210	243	247	155	X	X
2 + UMA 200 55/21	1117	787	2127	210	247	247	257	X	X
3 + UMA 200 90/21	-	943	2683	210	246	246	347	X	-
4 + UMA 250 110/21	-	1139	2668	210	249	253	442	X	-
5 + UMA 250 132/21	-	1295	2954	210	253	253	505	X	-
6 + UMA 250 160/21	-	1451	3220	210	253	270	561	X	-

Table 18: Technical data, 60 Hz / 460 V 3~, n = 3500 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 250 - 140 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m ³ /h	Maximum rated power	Rated power	Rated current	Efficiency	Number × cross-section of cores ¹²⁾	
	H ₀	P ₂	P _N	I _N	η _M	DOL	Y - Δ
	[m]	[kW]	[kW]	[A]	[%]	[mm ²]	[mm ²]
1 + UMA 150 30/21	60	30	36	67	85,5	F4 × 6	F3/F4 × 4
2 + UMA 200 55/21	121	61	66	110	86,5	F4 × 16	F3/F4 × 6
3 + UMA 200 90/21	181	91	108	179	88,0	F3/F4 × 16 ¹³⁾	F3/F4 × 16
4 + UMA 250 110/21	241	122	132	219	88,5	F3/F4 × 16 ¹³⁾	F3/F4 × 25
5 + UMA 250 132/21	302	152	158	289	89,0	F3/F4 × 25 ¹³⁾	F3/F4 × 25
6 + UMA 250 160/21	362	183	192	310	89,5	F3/F4 × 25 II	F3/R4 × 35

Table 19: Explanation of abbreviations

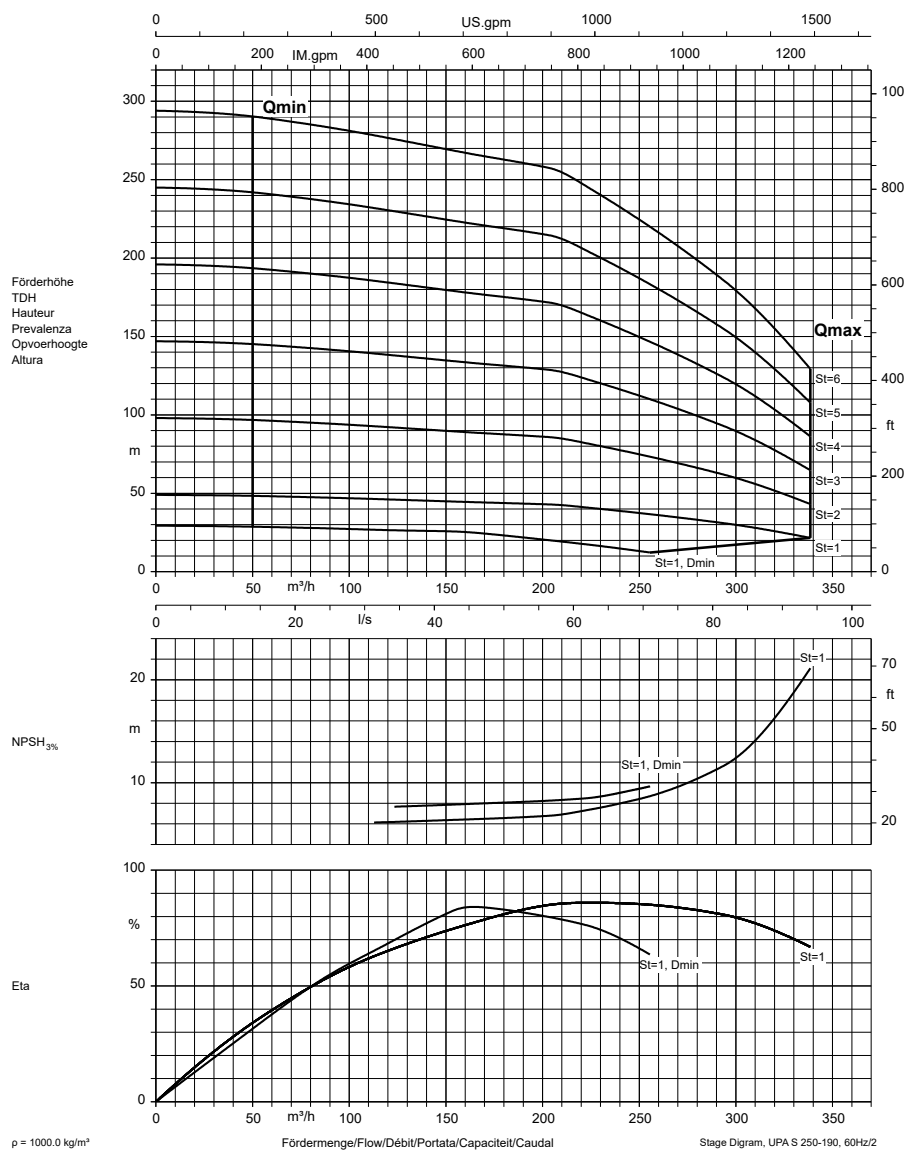
Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

11 Dimension indicated refers to version with standard pump end G6.

12 Designed for submerged use, 400 V, ≤ +30 °C

13 Delta configuration in cable connector with extension cable or in control cabinet

UPA S 250 - 190 / ..., number of stages 1 - 6, 60 Hz, n = 3500 rpm



Connection types

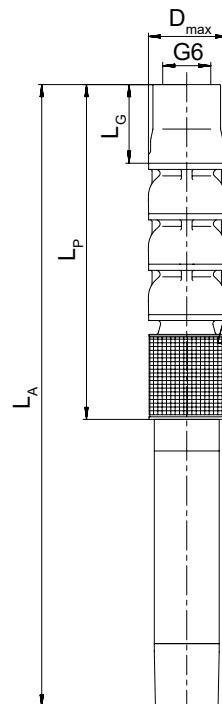
Standard pump end = G6

Alternative pump ends:

- G8
- DN 150, DN 200
- NPT6-8, NPT8-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 36)



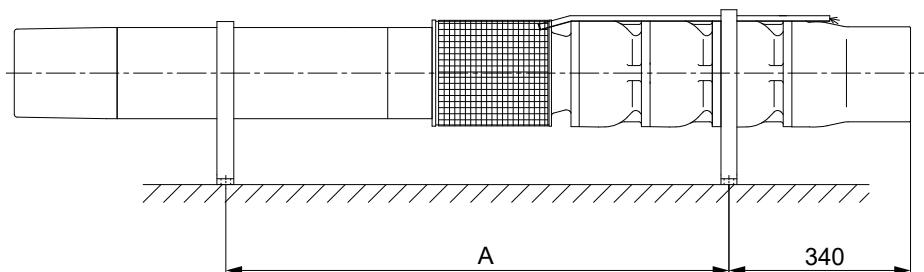
Operating range

$Q_{\text{min}} = 50 \text{ m}^3/\text{h}$

$Q_{\text{max}} = \text{end of stage curve}$

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 5)



Dimensions of UPA S 250 - 190 / ... [mm]

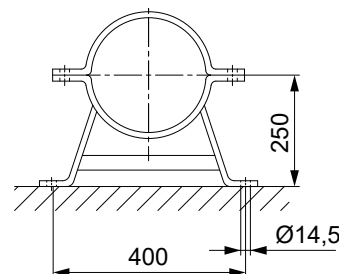


Table 20: Dimensions, weights and installation type depending on the motor, 60 Hz / 460 V 3~

UPA S 250 - 190 / ...	A	L _p ¹⁴⁾	L _A ¹⁴⁾	L _G ¹⁴⁾	D _{max} ¹⁴⁾		Total weight	Type of installation	
					DOL	Y - Δ		Vertical	Horizontal
	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]		
1 + UMA 150 30/21	900	643	1837	210	243	247	155	X	X
2 + UMA 200 55/21	1141	811	2151	210	247	247	258	X	X
3 + UMA 200 90/21	-	979	2719	210	246	246	349	X	-
4 + UMA 250 110/21	-	1187	2716	210	249	253	444	X	-
5 + UMA 250 132/21	-	1355	3014	210	253	253	508	X	-
6 + UMA 250 160/21	-	1523	3292	210	253	270	565	X	-

Table 21: Technical data, 60 Hz / 460 V 3~, n = 3500 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 250 - 190 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m ³ /h	Maximum rated power	Rated power	Rated current	Efficiency	Number × cross-section of cores ¹⁵⁾	
	H ₀ [m]	P ₂ [kW]	P _N [kW]	I _N [A]	η _M [%]	DOL [mm ²]	Y - Δ [mm ²]
1 + UMA 150 30/21	49	31	36	67	85,5	F4 × 6	F3/F4 × 4
2 + UMA 200 55/21	98	61	66	110	86,5	F4 × 16	F3/F4 × 6
3 + UMA 200 90/21	147	92	108	179	88,0	F3/F4 × 16 ¹⁶⁾	F3/F4 × 16
4 + UMA 250 110/21	196	123	132	219	88,5	F3/F4 × 16 ¹⁶⁾	F3/F4 × 25
5 + UMA 250 132/21	245	153	158	289	89,0	F3/F4 × 25 ¹⁶⁾	F3/F4 × 25
6 + UMA 250 160/21	294	184	192	310	89,5	F3/F4 × 25 II	F3/R4 × 35

Table 22: Explanation of abbreviations

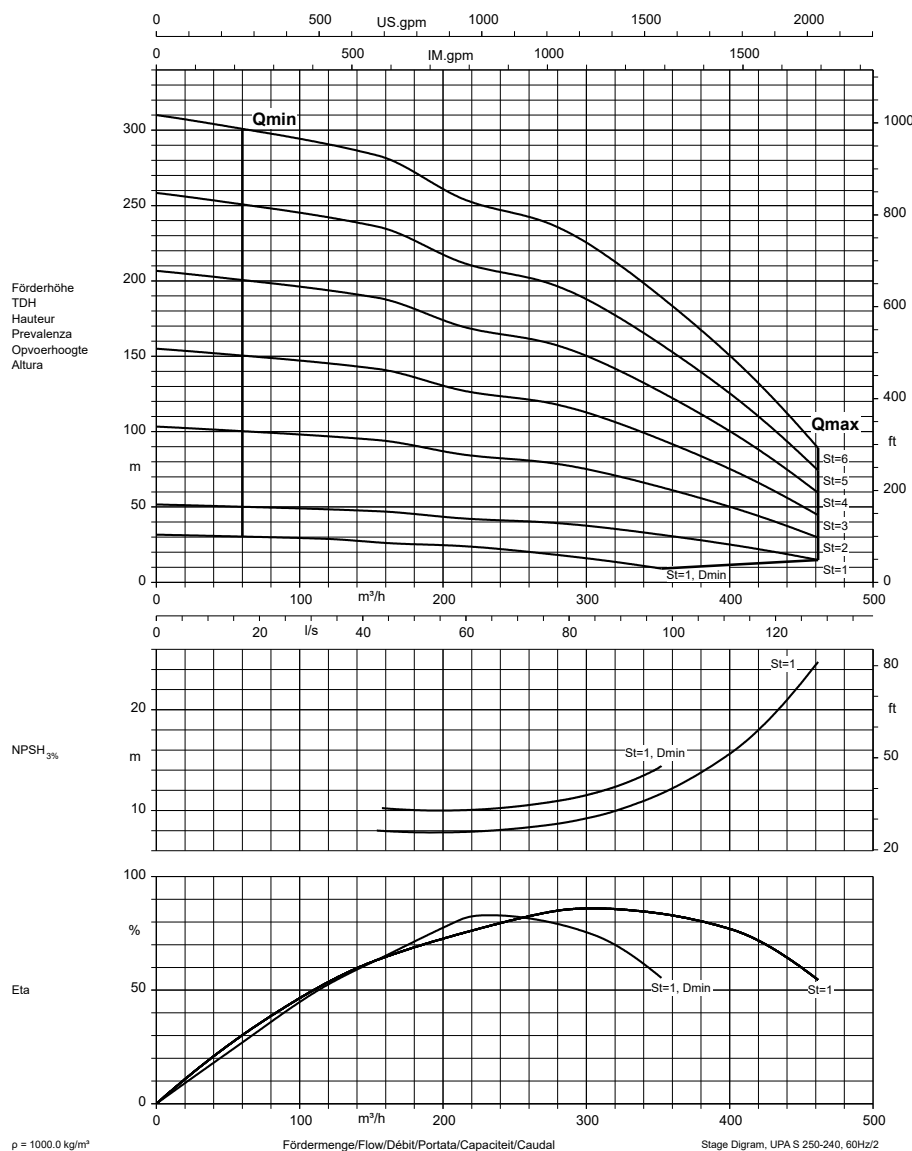
Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

14 Dimension indicated refers to version with standard pump end G6.

15 Designed for submerged use, 400 V, ≤ +30 °C

16 Delta configuration in cable connector with extension cable or in control cabinet

UPA S 250 - 240 / ..., number of stages 1 - 6, 60 Hz, n = 3500 rpm



Connection types

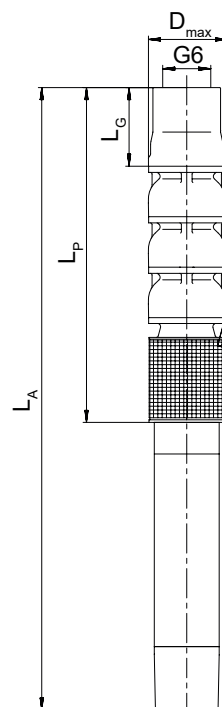
Standard pump end = G6

Alternative pump ends:

- G8
- DN 150, DN 200
- NPT6-8, NPT8-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 37)



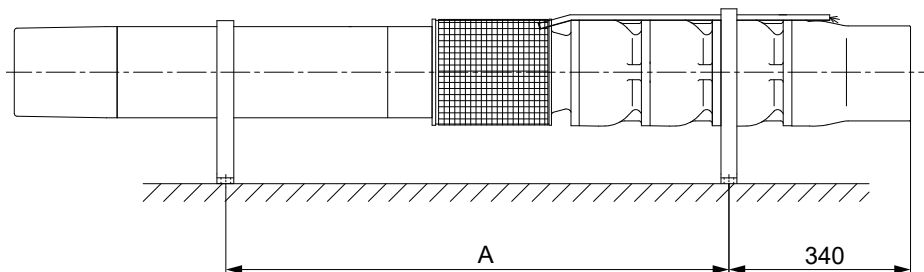
Operating range

$Q_{min} = 60 \text{ m}^3/\text{h}$

$Q_{max} = \text{end of stage curve}$

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 6)



Dimensions of UPA S 250 - 240 / ... [mm]

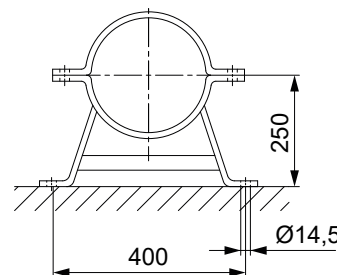


Table 23: Dimensions, weights and installation type depending on the motor, 60 Hz / 460 V 3~

UPA S 250 - 240 / ...	A	L _p ¹⁷⁾	L _A ¹⁷⁾	L _G ¹⁷⁾	D _{max} ¹⁷⁾		Total weight	Type of installation	
					DOL	Y - Δ		Vertical	Horizontal
	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]		
1 + UMA 150 37/22	-	648	1922	210	246	246	161	X	-
2 + UMA 200 65/21	1216	821	2291	210	246	246	279	X	X
3 + UMA 200 90/21	-	994	2734	210	245	245	346	X	-
3 + UMA 250 110/21	1459	1034	2563	210	248	253	421	X	X
4 + UMA 250 132/21	-	1207	2866	210	253	253	484	X	-
5 + UMA 250 160/21	-	1380	3149	210	253	270	540	X	-
6 + UMA 250 190/21	-	1553	3472	210	270	270	610	X	-

Table 24: Technical data, 60 Hz / 460 V 3~, n = 3500 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 250 - 240 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m ³ /h	Maximum rated power	Rated power	Rated current	Efficiency	Number × cross-section of cores ¹⁸⁾	
	H ₀ [m]	P ₂ [kW]	P _N [kW]	I _N [A]	η _M [%]	DOL [mm ²]	Y - Δ [mm ²]
1 + UMA 150 37/22	52	36	44	82	84,5	F3/F4 × 4 II	F3/F4 × 4
2 + UMA 200 65/21	103	72	77	128	87,5	F3/F4 × 10 ¹⁹⁾	F3/F4 × 10
3 + UMA 200 90/21	155	109	108	179	88,0	F3/F4 × 16 ¹⁹⁾	F3/F4 × 16
3 + UMA 250 110/21	155	109	132	219	88,5	F3/F4 × 16 ¹⁹⁾	F3/F4 × 25
4 + UMA 250 132/21	207	145	158	289	89,0	F3/F4 × 25 ¹⁹⁾	F3/F4 × 25
5 + UMA 250 160/21	258	181	192	310	89,5	F3/F4 × 25 II	F3/R4 × 35
6 + UMA 250 190/21	310	217	228	371	90,0	F3/R4 × 35 II	F3/R4 × 35

Table 25: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

17 Dimension indicated refers to version with standard pump end G6.

18 Designed for submerged use, 400 V, ≤ +30 °C

19 Delta configuration in cable connector with extension cable or in control cabinet

Technical data, 100 Hz

Tolerance to ISO 9906 Cl. 2B

The characteristic curves shown are valid for:

- Water at 20 °C
- Density 998 kg/m³
- Viscosity 1.00 mm²/s

Information on characteristic curves

The characteristic curves shown refer to pump sets driven by a UMA-S synchronous motor, n = 3000 rpm.

The characteristic curves shown are intended to allow pre-selection. Refer to the quotation for the precise selection data.

Selection chart UPA S 250, 100 Hz, n = 3000 rpm

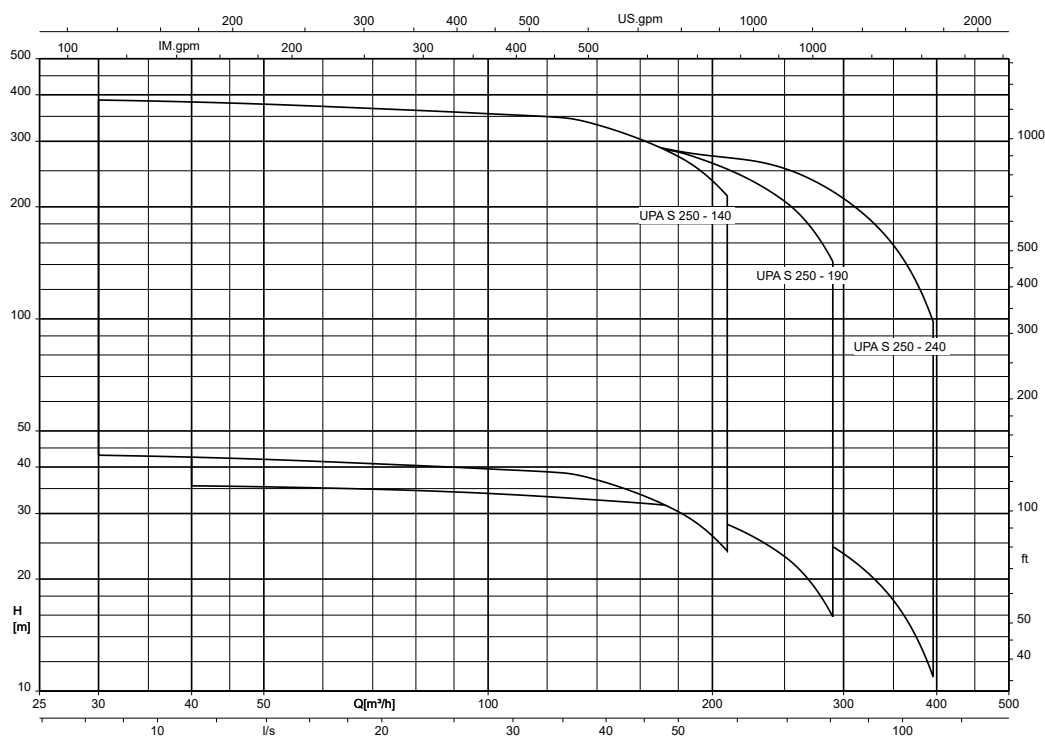
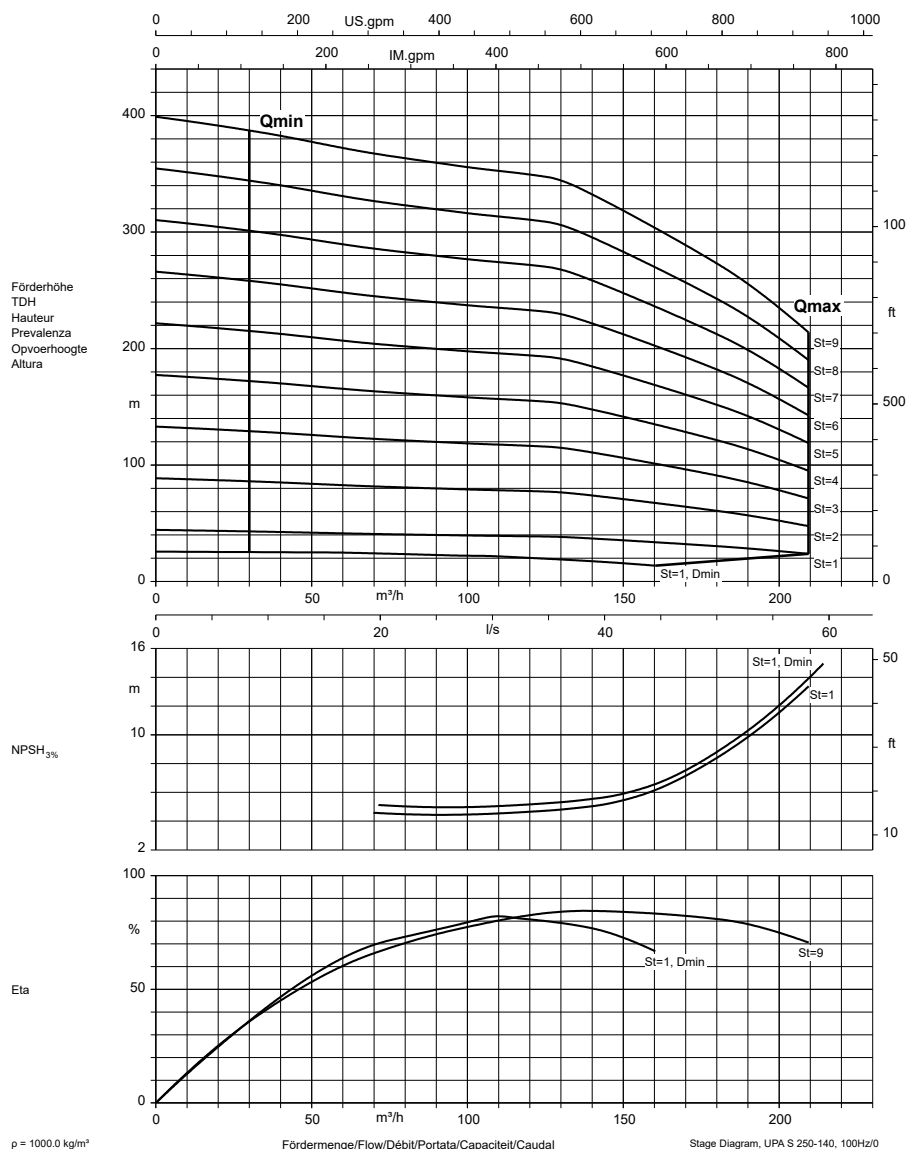


Fig. 3: Selection chart UPA S 250, 100 Hz

UPA S 250 - 140 / ..., number of stages 1 - 9, 100 Hz, n = 3000 rpm



Connection types

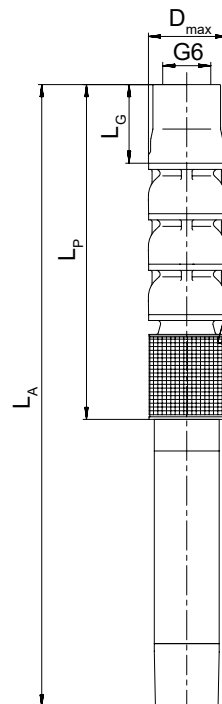
Standard pump end = G6

Alternative pump ends:

- G8
- DN 150, DN 200
- NPT6-8, NPT8-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 35)



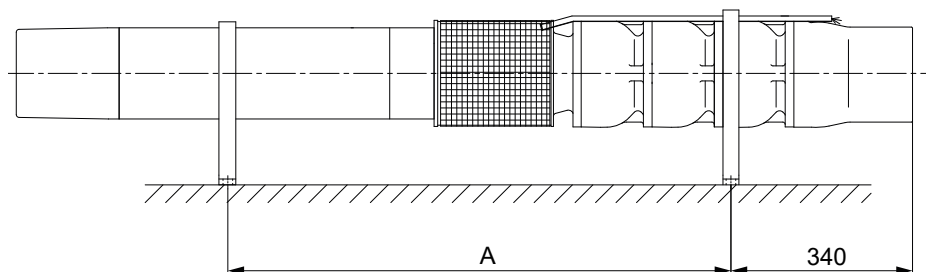
Operating range

$Q_{\text{min}} = 30 \text{ m}^3/\text{h}$

$Q_{\text{max}} = \text{end of stage curve}$

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 4)



Dimensions of UPA S 250 - 140 / ... [mm]

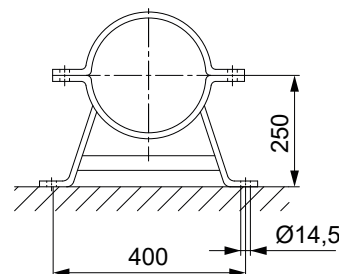


Table 26: Dimensions, weights and installation type depending on the motor, 100 Hz / 400 V 3~

UPA S 250 - 140 / ...	A	L _p ²⁰⁾	L _A ²⁰⁾	L _G ²⁰⁾	D _{max} ²⁰⁾		Total weight	Type of installation	
					VFD	VFD parallel	Material variant 1.4408 / 1.4517	Vertical	Horizontal
	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[kg]		
1 + UMA-S 150 37/42	777	631	1602	210	243	247	132	X	X
2 + UMA-S 200 75/42	1050	787	1992	210	242	246	231	X	X
3 + UMA-S 200 75/42	1206	943	2148	210	242	246	250	X	X
4 + UMA-S 200 100/42	1417	1099	2415	210	262	246	288	X	X
5 + UMA-S 200 130/42	-	1255	2739	210	-	246	335	X	-
6 + UMA-S 200 130/42	-	1411	2895	210	-	246	354	X	-
7 + UMA-S 250 200/42	-	1607	3266	210	-	270	481	X	-
8 + UMA-S 250 200/42	-	1763	3422	210	-	270	500	X	-
9 + UMA-S 250 200/42	-	1919	3578	210	-	270	519	X	-

Table 27: Technical data, 100 Hz / 400 V 3~, n = 3000 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 250 - 140 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m³/h	Maximum rated power	Rated power	Rated current	Efficiency	Number × cross-section of cores ²¹⁾	
	H ₀ [m]	P ₂ [kW]	P _N [kW]	I _N [A]	η _M [%]	VFD [mm²]	VFD parallel [mm²]
1 + UMA-S 150E 37/42	44	19	22	39	92,7	F4 × 6	F3/F4 × 4 II
2 + UMA-S 200D 75/42	89	38	45	74	93,3	F4 × 16	F3/F4 × 16 II
3 + UMA-S 200D 75/42	133	58	67	113	93,0	F4 × 16	F3/F4 × 16 II
4 + UMA-S 200D 100/42	177	77	83	146	93,3	R4 × 25	F3/F4 × 16 II
5 + UMA-S 200D 130/42	222	96	110	186	93,8	-	F3/F4 × 16 II
6 + UMA-S 200D 130/42	266	115	130	224	93,3	-	F3/F4 × 16 II
7 + UMA-S 250D 200/42	310	134	185	358	93,9	-	F3/R4 × 35 II
8 + UMA-S 250D 200/42	355	154	185	358	93,9	-	F3/R4 × 35 II
9 + UMA-S 250D 200/42	399	173	185	358	93,9	-	F3/R4 × 35 II

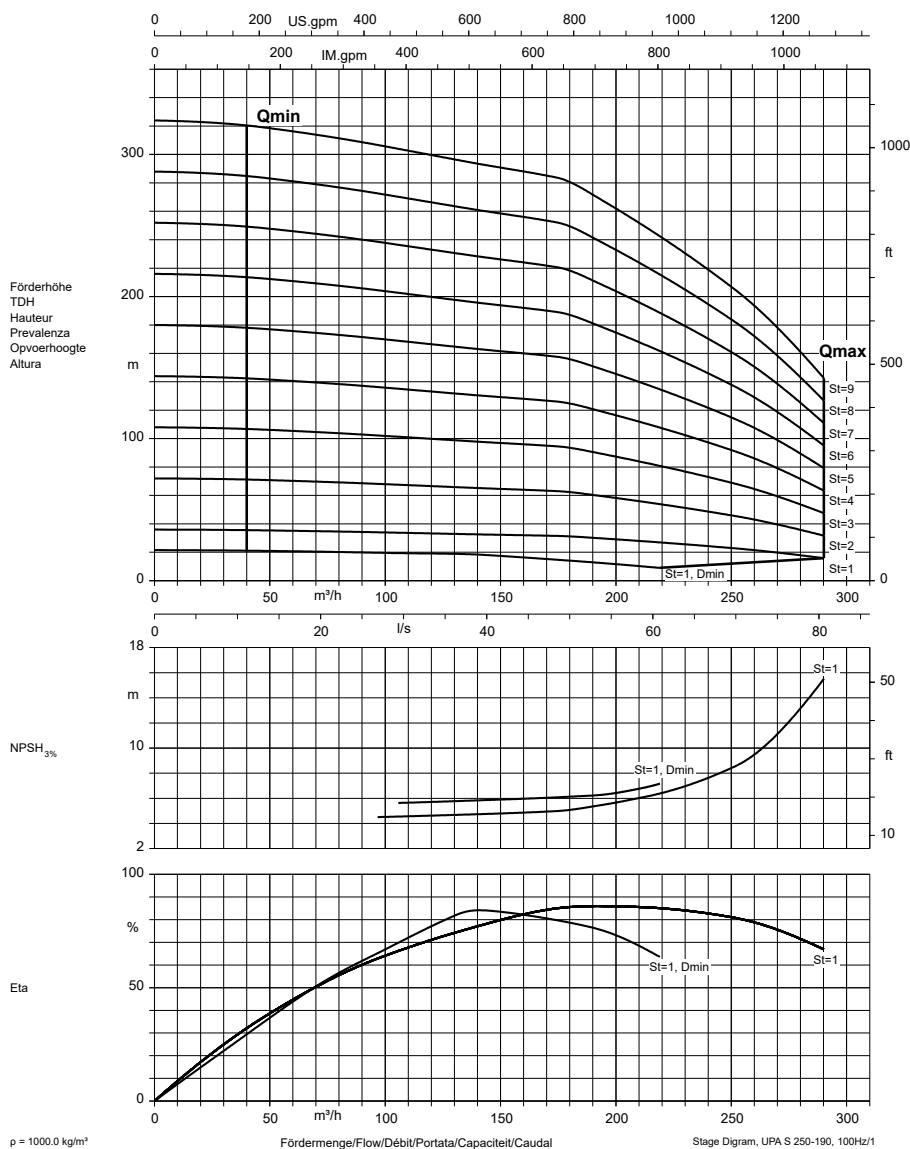
Table 28: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
R4	1 × 4-core, cable cross-section: round
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

20 Dimension indicated refers to version with standard pump end G6.

21 Designed for submerged use, 400 V, ≤ +30 °C

UPA S 250 - 190 / ..., number of stages 1 - 9, 100 Hz, n = 3000 rpm



Connection types

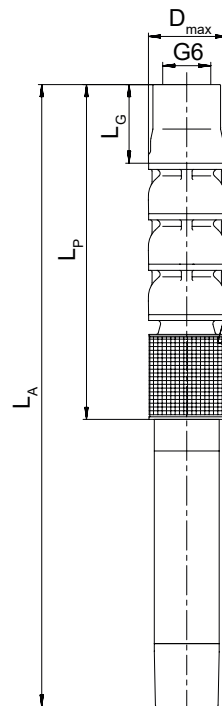
Standard pump end = G6

Alternative pump ends:

- G8
- DN 150, DN 200
- NPT6-8, NPT8-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 36)



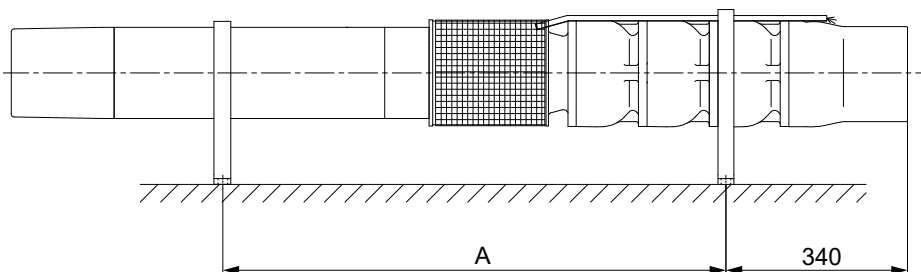
Operating range

$Q_{min} = 40 \text{ m}^3/\text{h}$

Q_{max} = end of stage curve

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 5)



Dimensions of UPA S 250 - 190 / ... [mm]

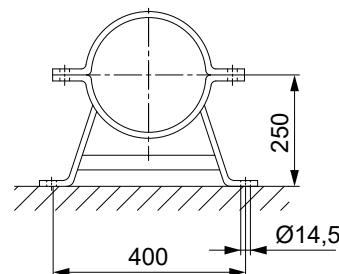


Table 29: Dimensions, weights and installation type depending on the motor, 100 Hz / 400 V 3~

UPA S 250 - 190 / ...	A	L _p ²²⁾	L _A ²²⁾	L _G ²²⁾	D _{max} ²²⁾		Total weight	Type of installation	
					VFD	VFD parallel	Material variant 1.4408 / 1.4517	Vertical	Horizontal
	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[kg]		
1 + UMA-S 150 37/42	789	643	1614	210	243	247	132	X	X
2 + UMA-S 200 75/42	1074	811	2016	210	242	246	232	X	X
3 + UMA-S 200 75/42	1242	979	2184	210	242	246	252	X	X
4 + UMA-S 200 100/42	1465	1147	2463	210	262	246	291	X	X
5 + UMA-S 200 100/42	-	1315	2631	210	-	246	310	X	-
6 + UMA-S 200 130/42	-	1483	2967	210	-	246	358	X	-
7 + UMA-S 250 200/42	-	1691	3350	210	-	270	485	X	-
8 + UMA-S 250 200/42	-	1859	3518	210	-	270	506	X	-
9 + UMA-S 250 200/42	-	2027	3686	210	-	270	525	X	-

Table 30: Technical data, 100 Hz / 400 V 3~, n = 3000 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 250 - 190 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m ³ /h	Maximum rated power	Rated power	Rated current	Efficiency	Number × cross-section of cores ²³⁾	
	H ₀ [m]	P ₂ [kW]	P _N [kW]	I _N [A]	η _M [%]	VFD [mm ²]	VFD parallel [mm ²]
1 + UMA-S 150E 37/42	36	19	22	39	92,7	F4 x 6	F3/F4 x 4 II
2 + UMA-S 200D 75/42	72	39	45	74	93,3	F4 x 16	F3/F4 x 16 II
3 + UMA-S 200D 75/42	108	58	67	113	93,0	F4 x 16	F3/F4 x 16 II
4 + UMA-S 200D 100/42	144	77	83	146	93,3	R4 x 25	F3/F4 x 16 II
5 + UMA-S 200D 100/42	180	97	100	177	92,9	-	F3/F4 x 16 II
6 + UMA-S 200D 130/42	216	116	130	224	93,3	-	F3/F4 x 16 II
7 + UMA-S 250D 200/42	252	135	185	358	93,9	-	F3/R4 x 35 II
8 + UMA-S 250D 200/42	288	154	185	358	93,9	-	F3/R4 x 35 II
9 + UMA-S 250D 200/42	324	174	185	358	93,9	-	F3/R4 x 35 II

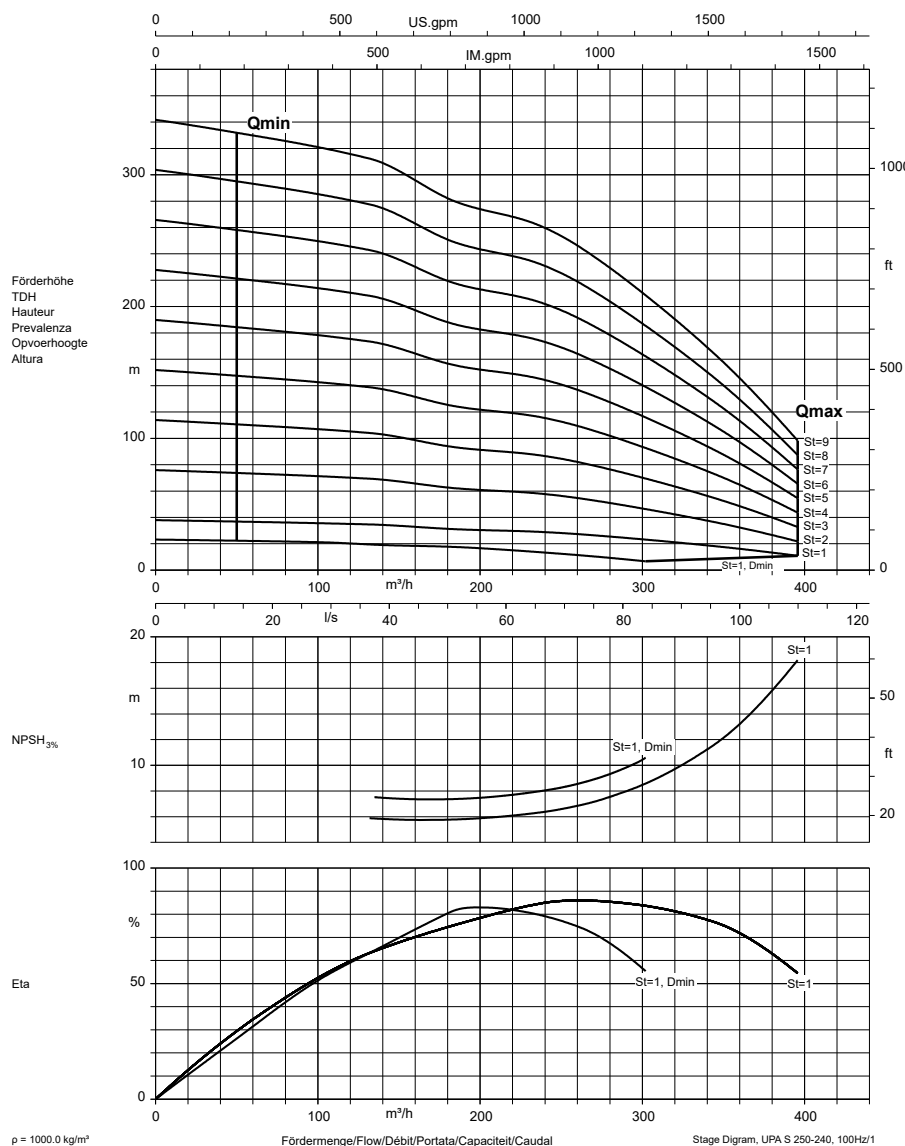
Table 31: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
R4	1 × 4-core, cable cross-section: round
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

22 Dimension indicated refers to version with standard pump end G6.

23 Designed for submerged use, 400 V, ≤ +30 °C

UPA S 250 - 240 / ..., number of stages 1 - 9, 100 Hz, n = 3000 rpm



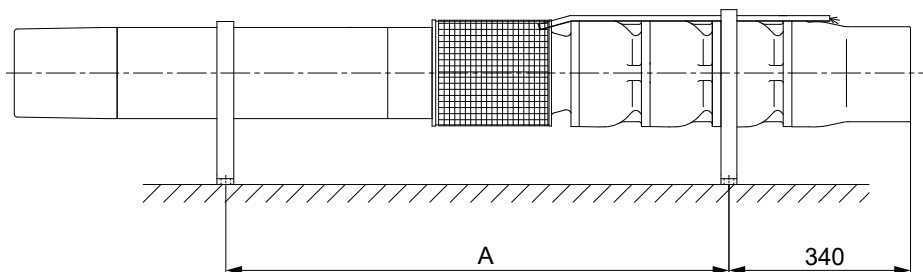
Operating range

$Q_{min} = 50 \text{ m}^3/\text{h}$

Q_{max} = end of stage curve

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 6)



Dimensions of UPA S 250 - 240 / ... [mm]

Connection types

Standard pump end = G6

Alternative pump ends:

- G8
- DN 150, DN 200
- NPT6-8, NPT8-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 37)

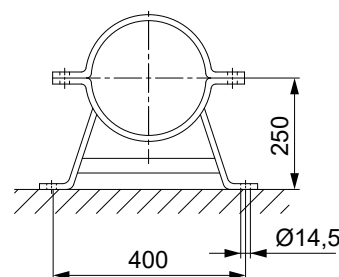
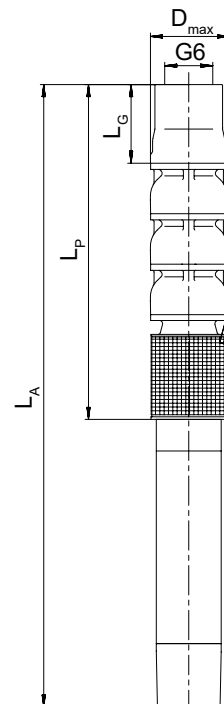


Table 32: Dimensions, weights and installation type depending on the motor, 100 Hz / 400 V 3~

UPA S 250 - 240 / ...	A	L _p ²⁴⁾	L _A ²⁴⁾	L _G ²⁴⁾	D _{max} ²⁴⁾		Total weight	Type of installation	
					VFD	VFD parallel		Vertical	Horizontal
					[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]		
1 + UMA-S 150 37/42	794	648	1619	210	242	246	131	X	X
2 + UMA-S 200 75/42	1084	821	2026	210	241	245	230	X	X
3 + UMA-S 200 75/42	1257	994	2199	210	-	245	249	X	X
4 + UMA-S 200 100/42	1485	1167	2483	210	-	245	287	X	X
5 + UMA-S 200 130/42	-	1340	2824	210	-	245	334	X	-
6 + UMA-S 250 200/42	-	1553	3212	210	-	270	460	X	-
7 + UMA-S 250 200/42	-	1726	3385	210	-	270	479	X	-
8 + UMA-S 250 200/42	-	1899	3558	210	-	270	498	X	-
9 + UMA-S 250 250/42	-	2072	3841	210	-	266	568	X	-

Table 33: Technical data, 100 Hz / 400 V 3~, n = 3000 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 250 - 240 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m ³ /h	Maximum rated power	Rated power	Rated current	Efficiency	Number × cross-section of cores ²⁵⁾	
						VFD	VFD parallel
						[mm ²]	[mm ²]
1 + UMA-S 150E 37/42	38	23	26	46	92,4	F4 x 6	F3/F4 x 4 II
2 + UMA-S 200D 75/42	76	46	55	91	93,3	F4 x 16	F3/F4 x 16 II
3 + UMA-S 200D 75/42	114	68	75	128	92,8	-	F3/F4 x 16 II
4 + UMA-S 200D 100/42	152	91	100	177	92,9	-	F3/F4 x 16 II
5 + UMA-S 200D 130/42	190	114	130	224	93,3	-	F3/F4 x 16 II
6 + UMA-S 250D 200/42	228	137	185	358	93,9	-	F3/R4 x 35 II
7 + UMA-S 250D 200/42	266	160	185	358	93,9	-	F3/R4 x 35 II
8 + UMA-S 250D 200/42	304	182	200	389	93,8	-	F3/R4 x 35 II
9 + UMA-S 250D 250/42	342	205	200	389	93,8	-	6xR1x35 II + E

Table 34: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
R1	1 × 1-core, cable cross-section: round
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

24 Dimension indicated refers to version with standard pump end G6.

25 Designed for submerged use, 400 V, ≤ +30 °C

Loss curves for lift check valves

Head losses caused by lift check valve for UPA S 250-140, pump end G6, DN150

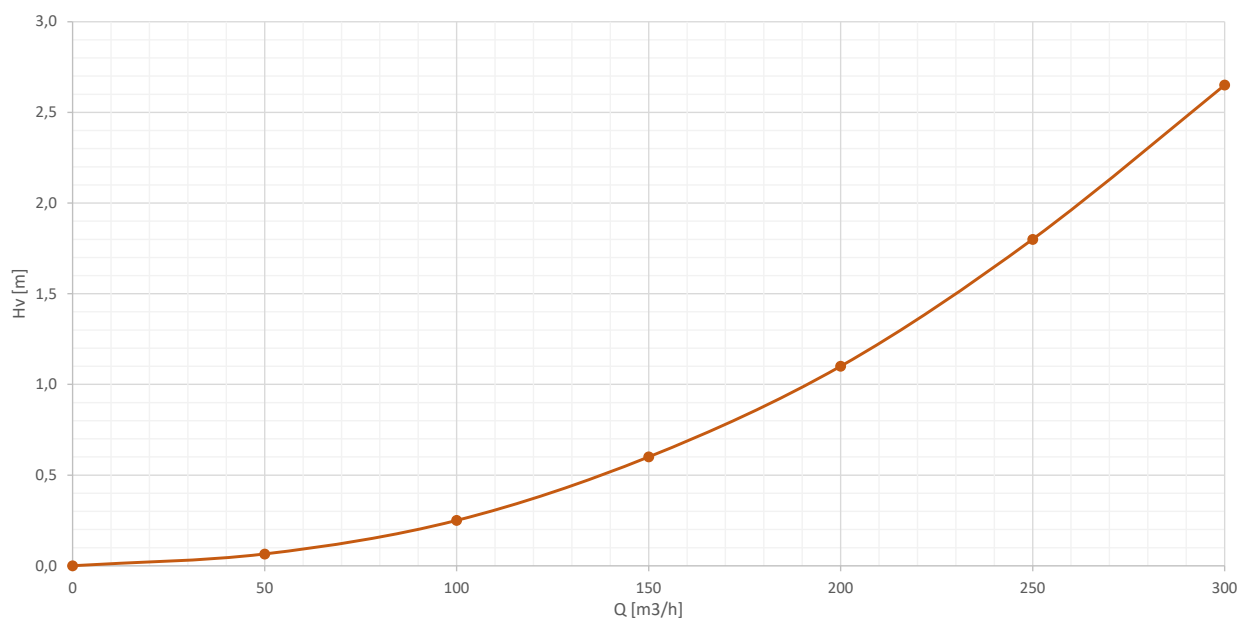


Fig. 4: Loss curve, UPA S 250-140

Head losses caused by lift check valve for UPA S 250-190, pump end G6, DN150

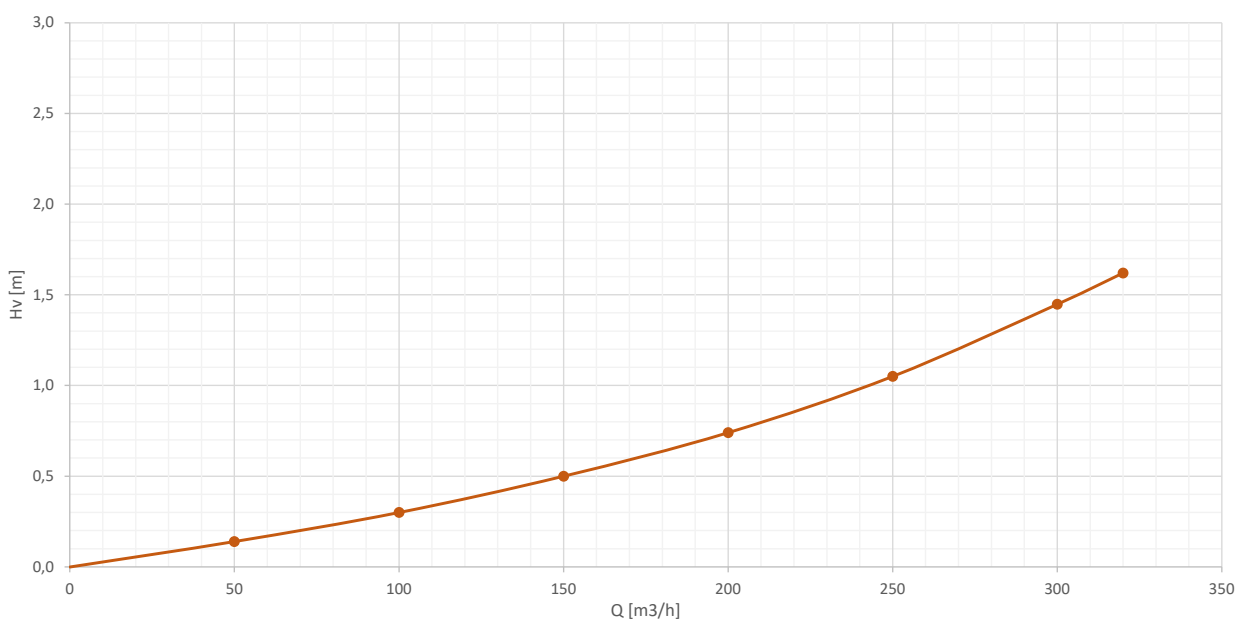


Fig. 5: Loss curve, UPA S 250-190

Head losses caused by lift check valve for UPA S 250-240, pump end G6, DN150

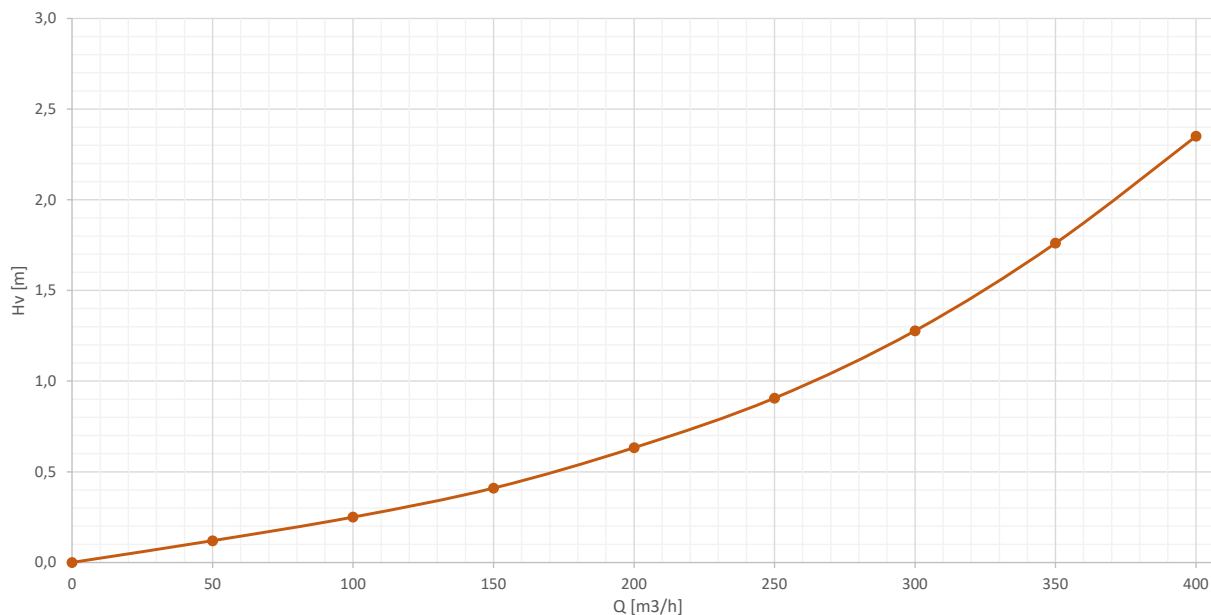


Fig. 6: Loss curve, UPA S 250-240

Head losses caused by lift check valve for UPA S 250-140, pump end G8, NPT 8, DN 200

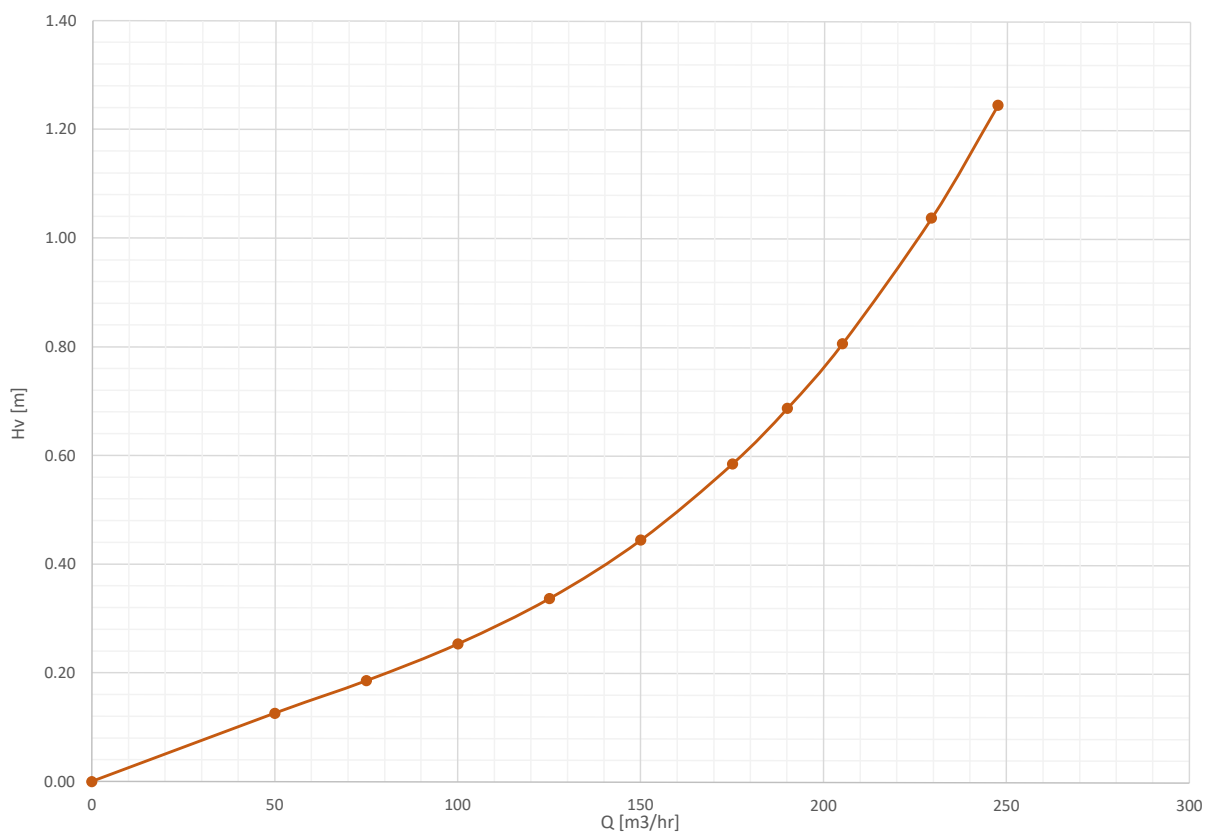


Fig. 7: Loss curve for UPA S 250-140

Head losses caused by lift check valve for UPA S 250-190, pump end G8, NPT 8, DN 200

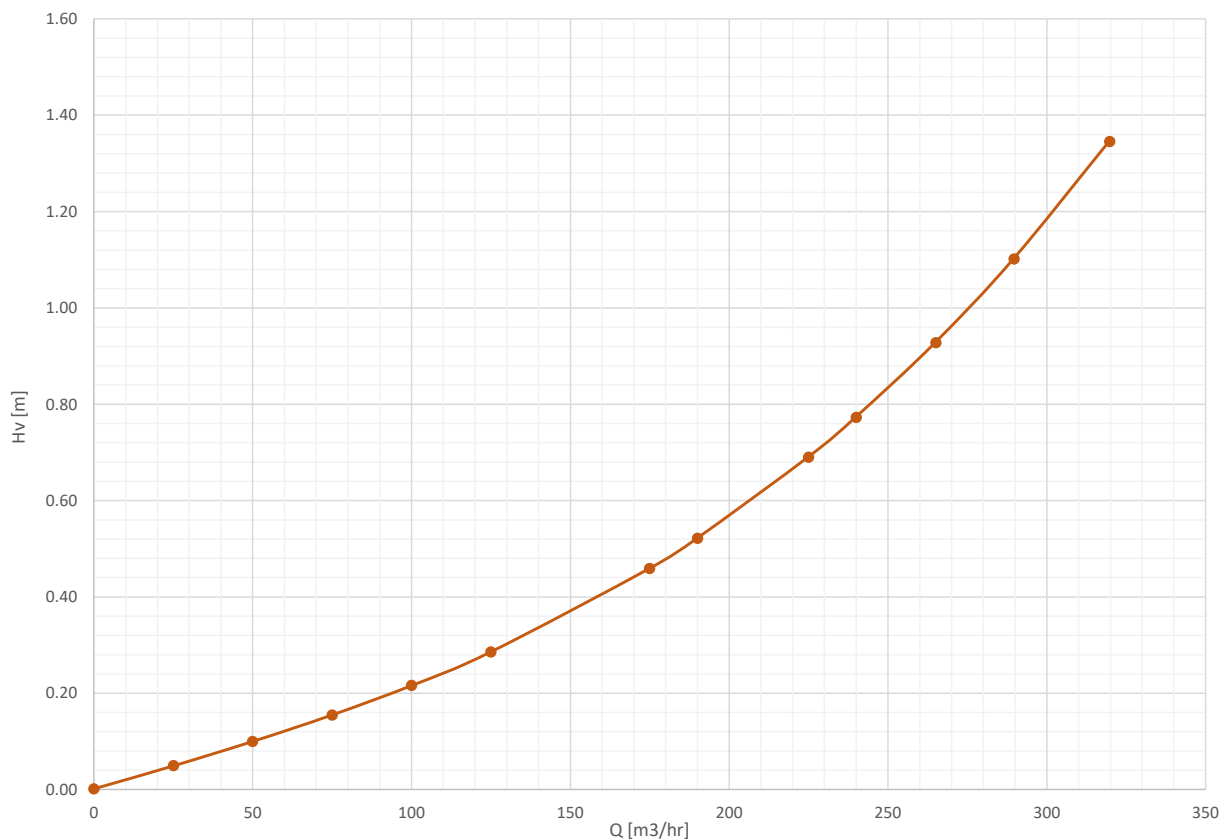


Fig. 8: Loss curve for UPA S 250-190

Head losses caused by lift check valve for UPA S 250-240, pump end G8, NPT 8, DN 200

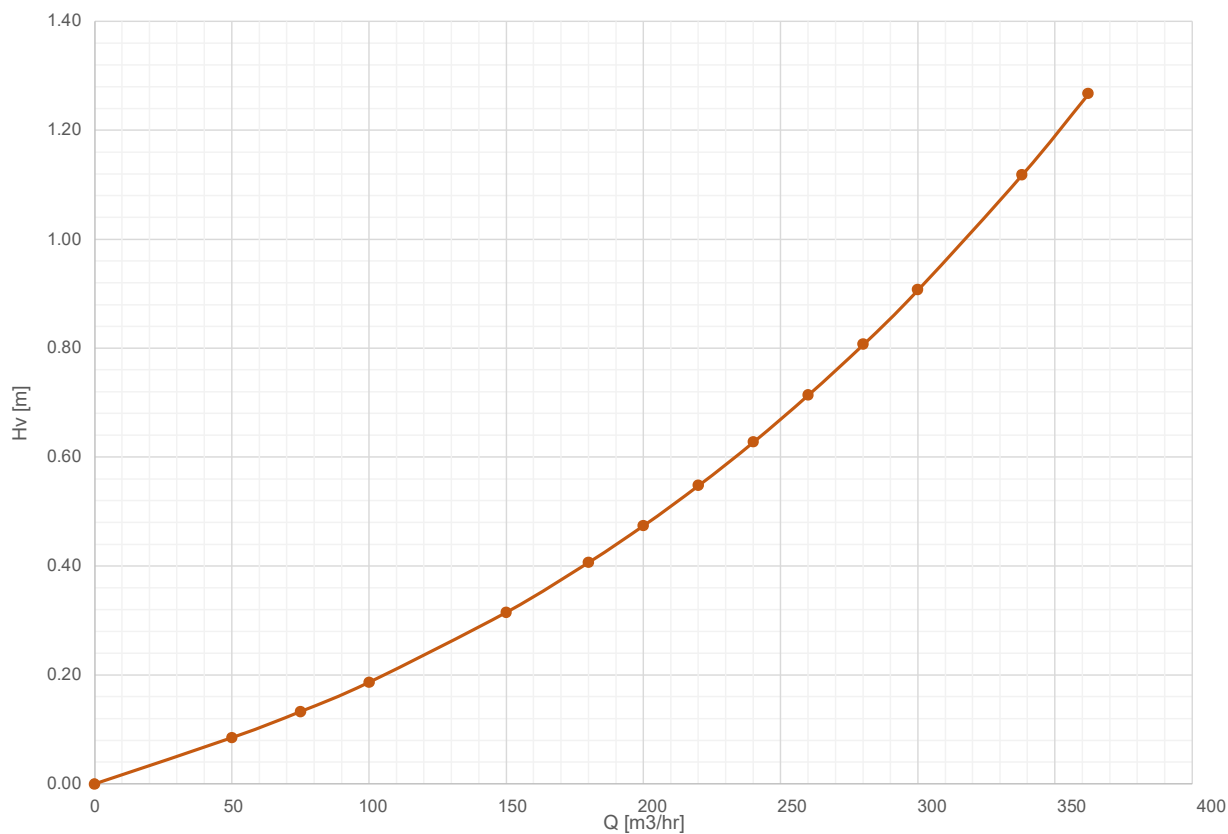


Fig. 9: Loss curve for UPA S 250-240

Dimensions of alternative pump end

i On versions with an alternative pump end, dimensions $D_{\max}^*$ and L_A^* differ from those with the standard pump end.

- The new dimension $D_{\max}^*$ for the selected size can be found in the following tables.
- The new dimension L_A^* must be calculated: $L_A^* = L_A - L_G + L_G^*$
 - Dimensions L_A und L_G for the selected size can be found in the standard "Dimensions" table.
 - Dimension L_G^* for the selected size can be found in the following tables.

Table 35: Dimensions, alternative pump ends, UPA S 250-140

Pump end		$D_{\max}^*$	L_G^*
		[mm]	[mm]
Lift check valve / Connection nozzle	DN 150 / PN10, PN16:	295	179
	DN 150 / PN 25, PN 40	303	179
	DN 150 / PN 63	345	179
Lift check valve	G8 / NPT8-8	297	600
	DN 200 / PN 10, PN 16	352	560
	DN 200 , PN 25	362	560
	DN 200 , PN 40	375	560
	DN 200 , PN 63	421	560
Connection nozzle	G8 / NPT8-8	297	525
	DN 200 / PN 10, PN 16	352	470
	DN 200 , PN 25	362	470
	DN 200 , PN 40	375	470
	DN 200 , PN 63	421	470

Table 36: Dimensions, alternative pump ends, UPA S 250-190

Pump end		$D_{\max}^*$	L_G^*
		[mm]	[mm]
Lift check valve / Connection nozzle	DN 150 / PN10, PN16:	295	179
	DN 150 / PN 25, PN 40	303	179
	DN 150 / PN 63	345	179
Lift check valve	G8 / NPT8-8	297	535
	DN 200 / PN 10, PN 16	352	495
	DN 200 , PN 25	362	495
	DN 200 , PN 40	375	495
	DN 200 , PN 63	421	495
Connection nozzle	G8 / NPT8-8	297	460
	DN 200 / PN 10, PN 16	352	405
	DN 200 , PN 25	362	405
	DN 200 , PN 40	375	405
	DN 200 , PN 63	421	405

Table 37: Dimensions, alternative pump ends, UPA S 250-240

Pump end		$D_{\max}^*$	L_G^*
		[mm]	[mm]
Lift check valve / Connection nozzle	DN 150 / PN10, PN16:	295	179
	DN 150 / PN 25, PN 40	303	179
	DN 150 / PN 63	345	179
Lift check valve	G8 / NPT8-8	306	535
	DN 200 / PN 10, PN 16	352	495
	DN 200 , PN 25	362	495
	DN 200 , PN 40	375	495
	DN 200 , PN 63	430	495
Connection nozzle	G8 / NPT8-8	306	460
	DN 200 / PN 10, PN 16	352	405
	DN 200 , PN 25	362	405
	DN 200 , PN 40	375	405
	DN 200 , PN 63	430	405

Scope of supply

Depending on the model, the following items are included in the scope of supply:

- Pump set with motor lead
- Back-up name plate

Optional:

- Extension cable connected or supplied but not fitted
- Cable connector
- Cable clips
- Pedestals

- Cooling shroud, suction shroud or pressure shroud
- Supporting clamps and mounting clamps
- Electrical protection equipment
- Automatic control units
- Centring device
- Spare parts
- Hydraulic performance test
- Frequency inverter
- Voltage filter
- Dry running protection device

Cross sectional drawings with list of components

UPA S 250 - 140/190/240, standard design

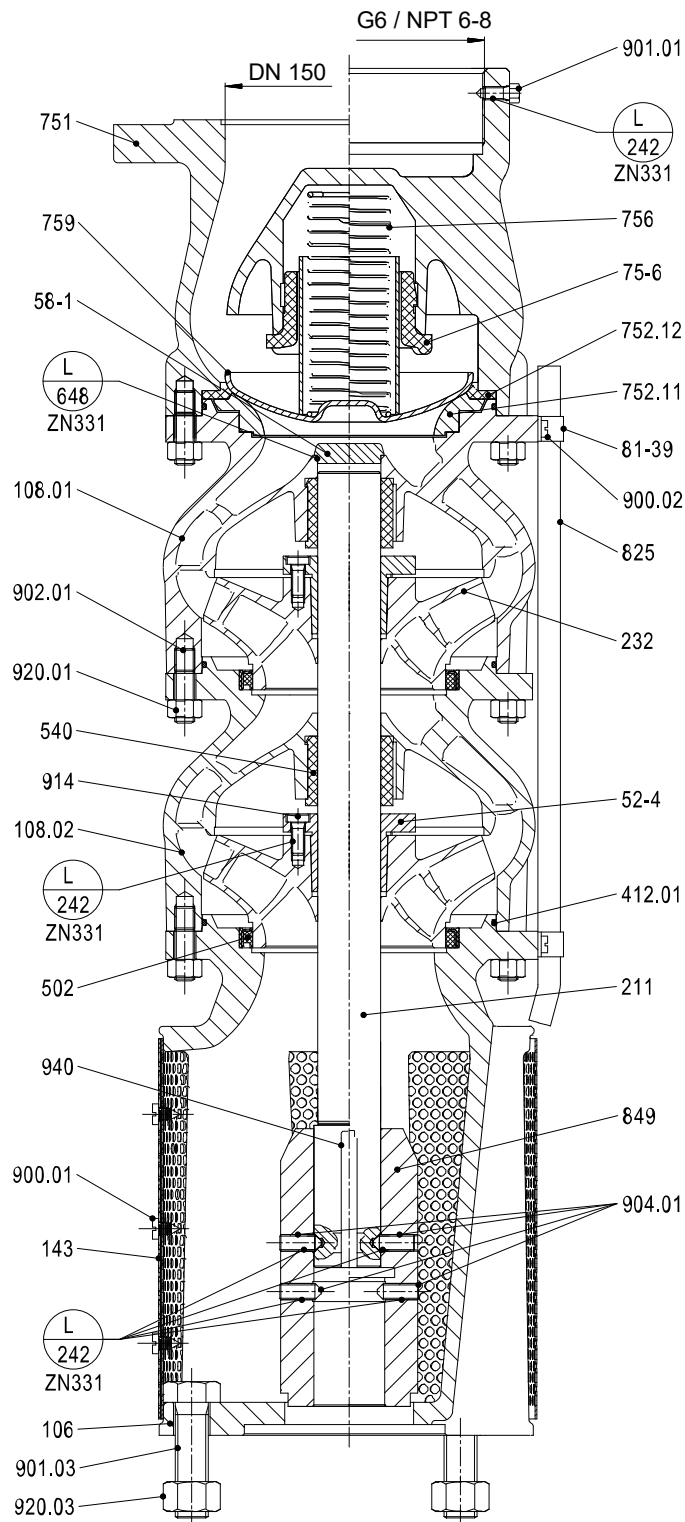


Fig. 10: UPA S 250 incl. lift check valve, for connection to UMA 250 or UMA-S 250 motor, standard design

3400.57/02-EN

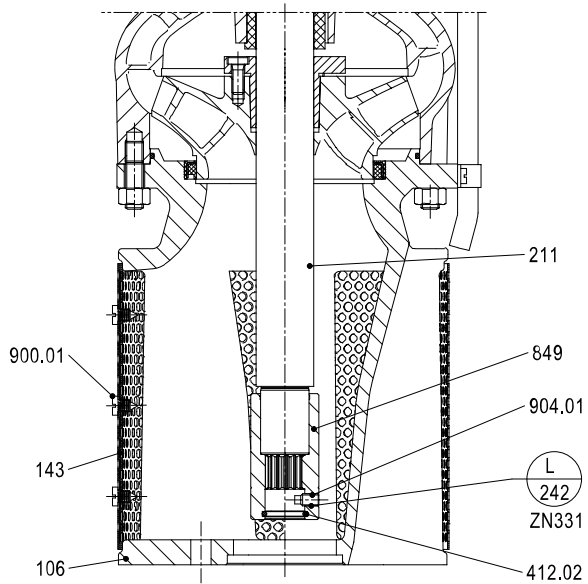


Fig. 11: UPA S 250 with connection for UMA 150 or UMA-S 150 motor, standard design

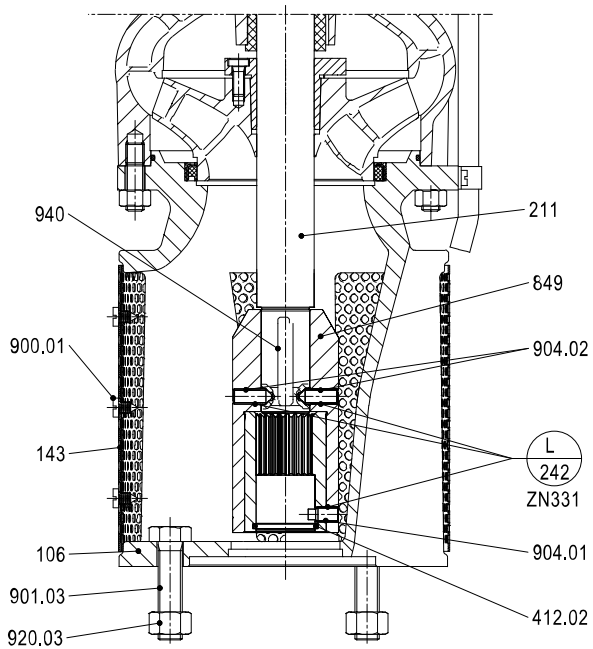


Fig. 12: UPA S 250 with connection for UMA 200 or UMA-S 200 motor, standard design

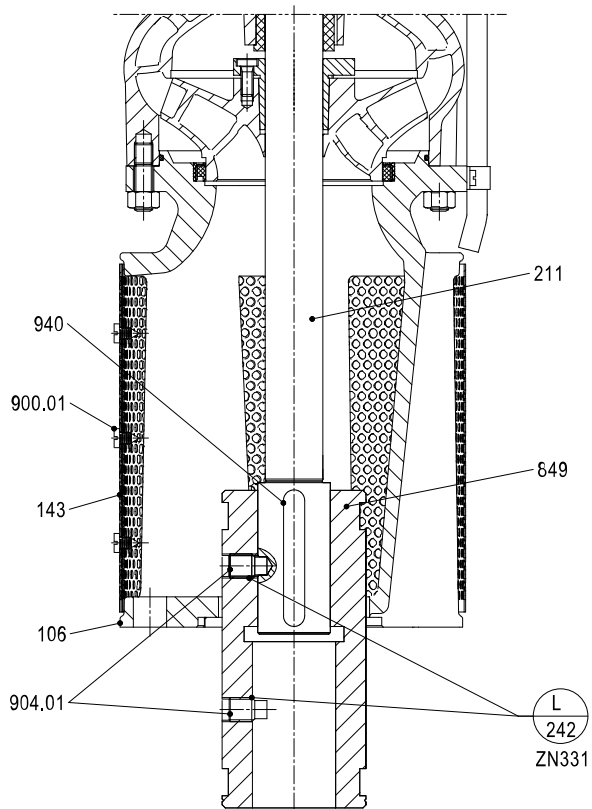


Fig. 13: UPA S 250 with connection for UMA 300 motor, standard design

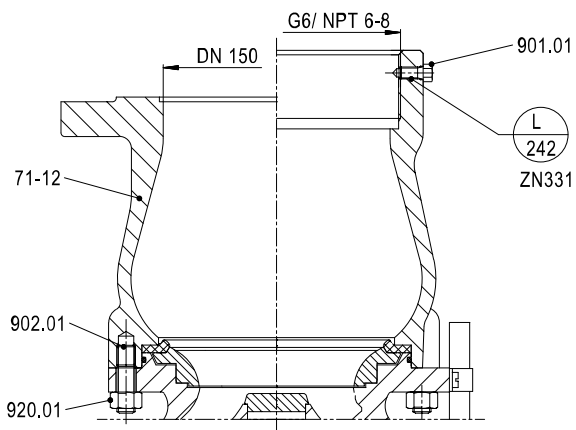


Fig. 14: UPA S 250, connection nozzle, standard design

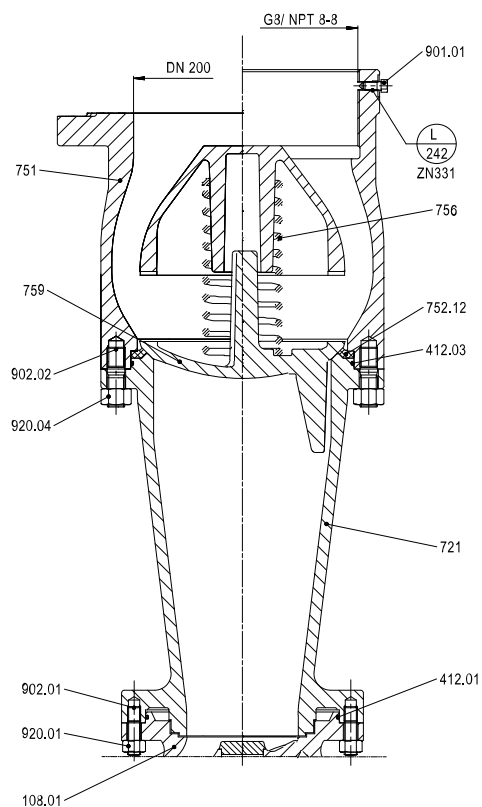


Fig. 15: UPA S 250, with valve body

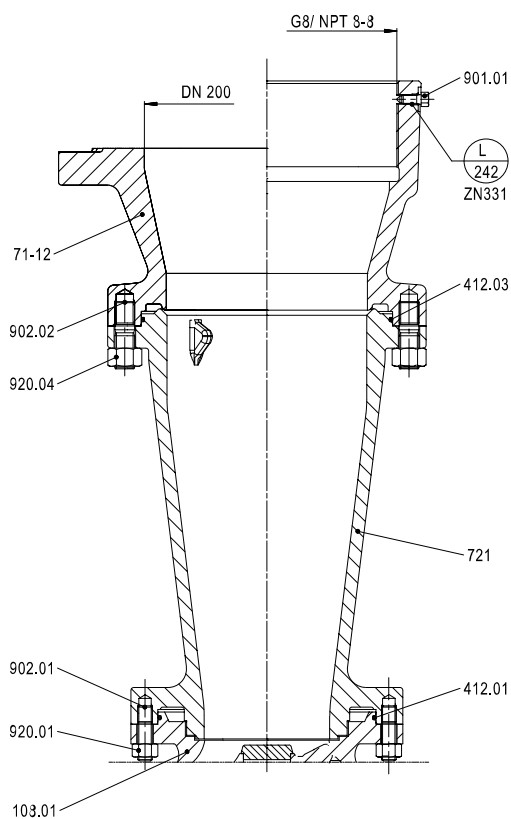


Fig. 16: UPA S 250, with connection nozzle

Table 38: Symbols key

Symbol	Description
	Always secure screw/bolt/stud connections marked with this symbol with Loctite 242 .
	Always secure connections marked with this symbol with Loctite 648 .

Table 39: List of components: UPA S 250 - 140/190/240, standard design

Quantity	Part No.	Description	Motor size				Scope of supply of spare parts kit
			6 inch UMA 150, UMA-S 150	8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
1 per stage	52-4 ²⁶⁾	Locking sleeve	✓	✓	✓	✓	52-4 + 914
1	58-1	Protecting plug	✓	✓	✓	✓	-
1	71-12 ²⁷⁾	Connection nozzle, flanged	✓	✓	✓	✓	71-12 + 902.01, 920.01
		Connection nozzle, threaded	✓	✓	✓	✓	71-12 + 901.01, 902.01, 920.01
1	71-12 ²⁸⁾	Connection nozzle, flanged	✓	✓	✓	✓	71-12 + 902.02, 920.04
		Connection nozzle, threaded	✓	✓	✓	✓	71-12 + 901.01, 902.02, 920.04
1	721	Adapter	✓	✓	✓	✓	721 + 412.03, 902.01, 920.01
1	75-6 ²⁶⁾²⁷⁾	Valve disc guide	✓	✓	✓	✓	-
2 per electric cable	81-39	Clamp	✓	✓	✓	✓	-
1	106	Suction casing	✓	-	-	✓	106 + 143, 412.01, 502, 900.01
			-	✓	✓	-	106 + 143, 412.01, 502, 900.01, 901.03, 920.03
1	108.01	Stage casing (last stage)	✓	✓	✓	✓	108.01 + 58-1, 412.01, 540, 902.01, 920.01
1 per stage -1	108.02	Stage casing	✓	✓	✓	✓	108.02 + 412.01, 502, 540, 902.01, 920.01
1	143 ²⁶⁾	Suction strainer	✓	✓	✓	✓	143 + 900.01
1	211	Pump shaft	✓	-	-	-	211 + 412.02, 540, 849, 904.01
			-	✓	-	-	211 + 412.02, 540, 849, 904.01, 904.02, 940
			-	-	✓	✓	211 + 540, 849, 904.01, 940
1 per stage	232	Clockwise impeller	✓	✓	✓	✓	232 + 52-4, 914
1 per stage +1	412.01 ²⁶⁾	O-ring	✓	✓	✓	✓	-
1	412.02 ²⁶⁾	O-ring	✓	✓	-	-	-
1	412.03 ²⁶⁾²⁸⁾	O-ring	✓	✓	✓	✓	-
1 per stage	502 ²⁶⁾	Casing wear ring	✓	✓	✓	✓	-
1 per stage	540 ²⁶⁾	Bush	✓	✓	✓	✓	-
1	751 ²⁷⁾	Valve body, flanged	✓	✓	✓	✓	751 + 75-6, 752.11, 752.12, 756, 759, 902.01, 920.01
		Valve body, threaded	✓	✓	✓	✓	751 + 75-6, 752.11, 752.12, 756, 759, 901.01, 902.01, 920.01
1	751 ²⁸⁾	Valve body, flanged	✓	✓	✓	✓	751 + 752.12, 756, 759, 902.02, 920.04
		Valve body, threaded	✓	✓	✓	✓	751 + 752.12, 756, 759, 901.01, 902.02, 920.04
1	752.11 ²⁷⁾	Valve seat (metal)	✓	✓	✓	✓	752.11 + 752.12
1	752.12 ²⁶⁾	Valve seat (EPDM)	✓	✓	✓	✓	-
1	756 ²⁶⁾	Valve spring	✓	✓	✓	✓	-
1	759	Valve disc	✓	✓	✓	✓	759 + 752.11 ²⁷⁾ , 75-6 ²⁷⁾ , 752.12, 756
1 per electric cable	825	Cable guard	✓	✓	✓	✓	-
1	849	Sleeve coupling	✓	-	-	-	849 + 412.02, 904.01
			-	✓	-	-	849 + 412.02, 904.01, 904.02, 940
			-	-	✓	✓	849 + 904.01, 940
3	900.01	Bolt/screw	✓	✓	✓	✓	-

26 Recommended spare part

27 Only for G6, NPT6-8 and DN 150 valve bodies

28 Only for G8, NPT8-8 and DN 200 valve bodies

Quantity	Part No.	Description	Motor size				Scope of supply of spare parts kit
			6 inch UMA 150, UMA-S 150	8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
4 per electric cable	900.02	Bolt/screw	✓	✓	✓	✓	-
2	901.01	Hexagon head bolt	✓	✓	✓	✓	-
4	901.03	Hexagon head bolt	-	✓	✓	-	-
(1 per stage +1) × 12	902.01	Stud	✓	✓	✓	✓	-
12	902.02	Stud	✓	✓	✓	✓	-
1	904.01	Grub screw	✓	✓	-	-	-
4			-	-	✓	-	-
2			-	-	-	✓	-
2	904.02	Grub screw	-	✓	-	-	-
1 per stage × 4	914	Hexagon socket head cap screw	✓	✓	✓	✓	-
(1 per stage +1) × 12	920.01	Nut	✓	✓	✓	✓	-
4	920.03	Nut	-	✓	✓	-	-
12	920.04	Nut	✓	✓	✓	✓	-
1	940 ²⁶⁾	Key	-	✓	✓	✓	-

UPA S 250 - 140/190/240, wear-resistant version

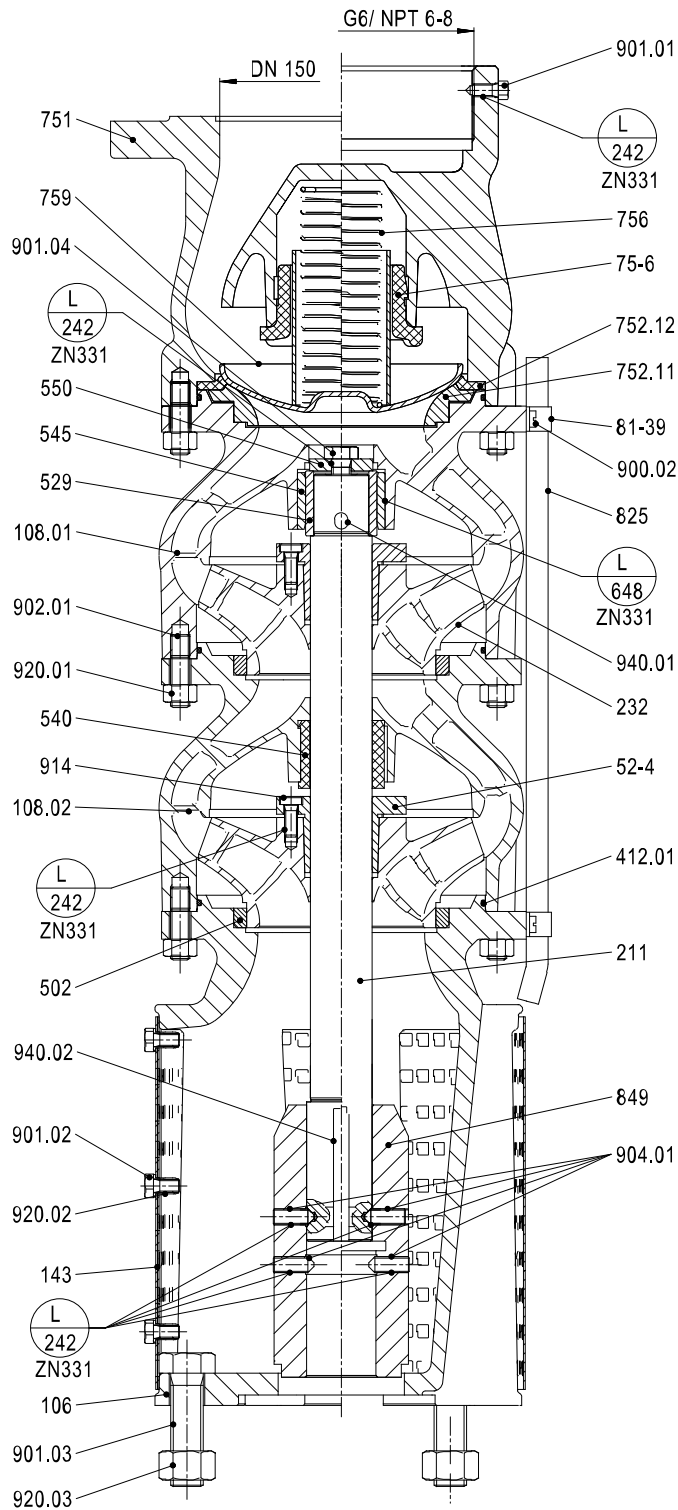


Fig. 17: UPA S 250 incl. lift check valve, for connection to UMA 250 or UMA-S 250 motor, wear-resistant version

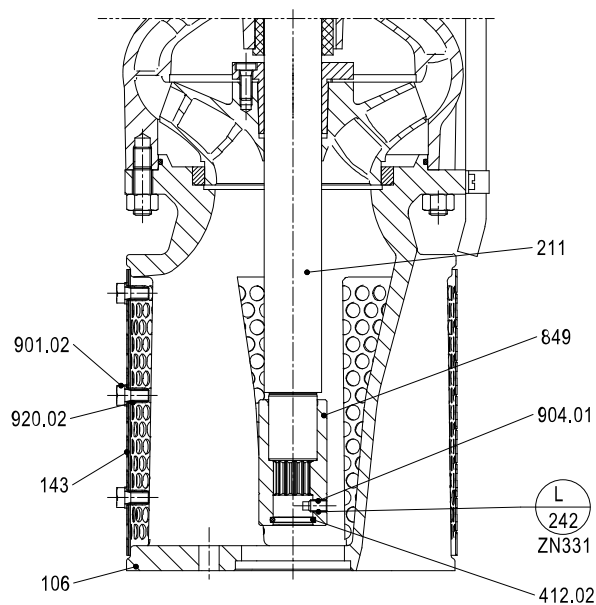


Fig. 18: UPA S 250 with connection for UMA 150 or UMA-S 150 motor, wear-resistant version

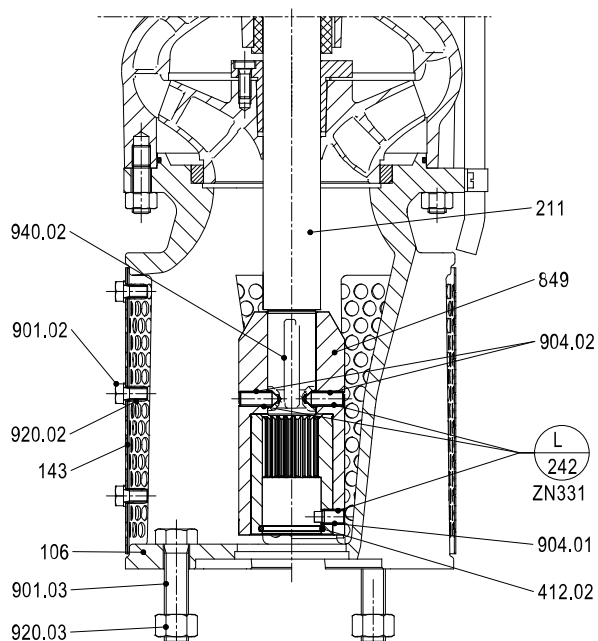


Fig. 19: UPA S 250 with connection for UMA 200 or UMA-S 200 motor, wear-resistant version

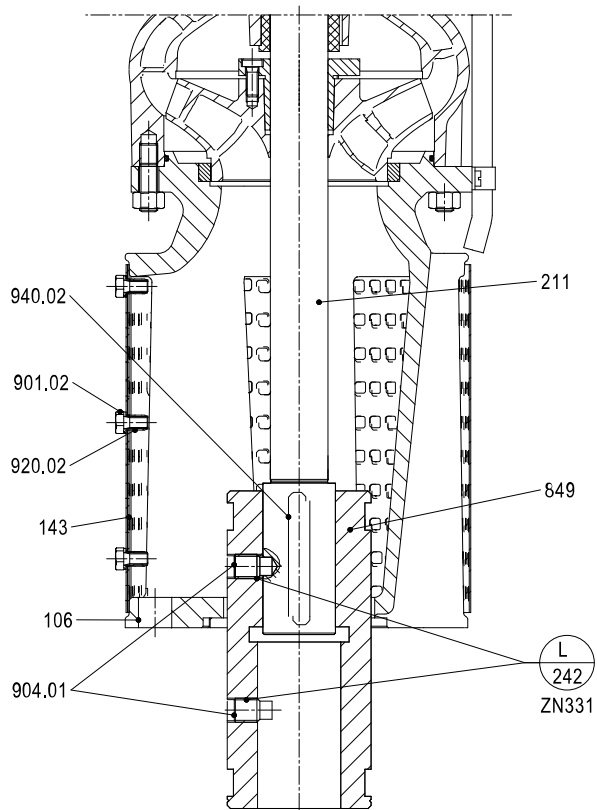


Fig. 20: UPA S 250 with connection for UMA 300 motor, wear-resistant version

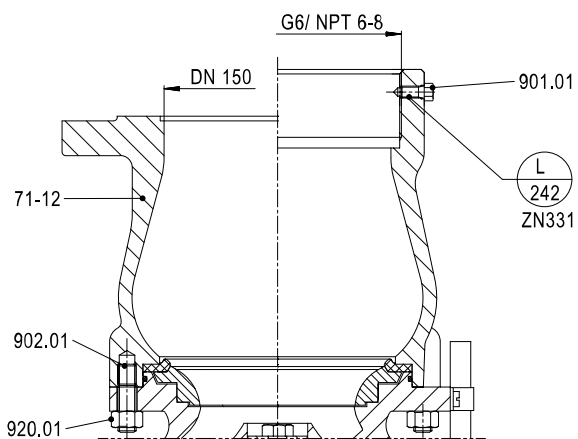


Fig. 21: UPA S 250, connection nozzle, wear-resistant version

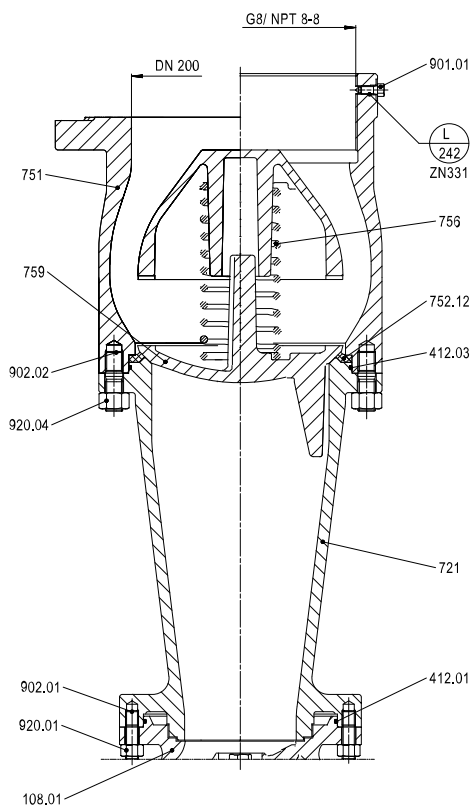


Fig. 22: UPA S 250, valve body, wear-resistant version

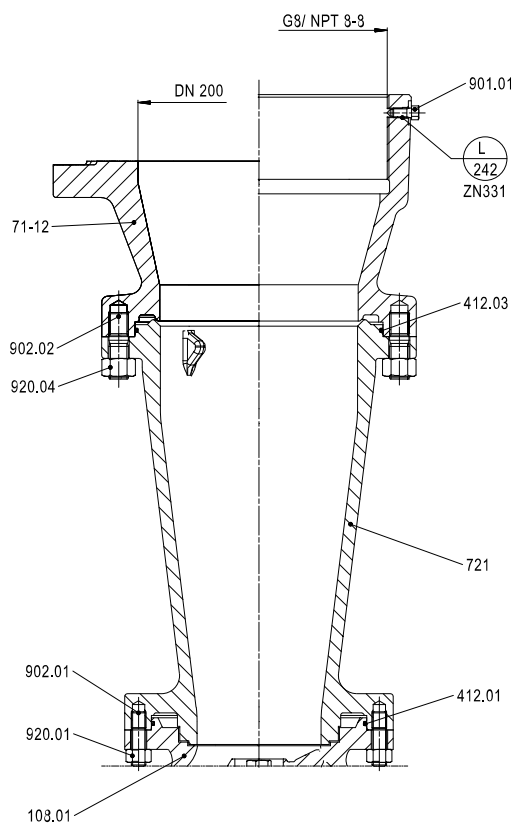


Fig. 23: UPA S 250, connection nozzle, wear-resistant version

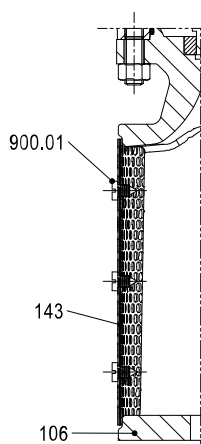


Fig. 24: UPA S 250, screen, wear-resistant version

Table 40: Symbols key

Symbol	Description
	Always secure screw/bolt/stud connections marked with this symbol with Loctite 242 .
	Always secure connections marked with this symbol with Loctite 648 .

Table 41: List of components: UPA S 250 - 140/190/240, wear-resistant version

Quantity	Part No.	Description	Motor size				Scope of supply of spare parts kit
			6 inch UMA 150, UMA-S 150	8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
1 per stage	52-4 ²⁹⁾	Locking sleeve	✓	✓	✓	✓	52-4 + 914
1	71-12 ³⁰⁾	Connection nozzle, flanged	✓	✓	✓	✓	71-12 + 902, 920.01
		Connection nozzle, threaded	✓	✓	✓	✓	71-12 + 901.01, 902, 920.01
1	71-12 ³¹⁾	Connection nozzle, flanged	✓	✓	✓	✓	71-12 + 902.02, 920.04
		Connection nozzle, threaded	✓	✓	✓	✓	71-12 + 901.01, 902.02, 920.04
1	721 ³¹⁾	Adapter	✓	✓	✓	✓	721 + 412.03, 902.01, 920.01
1	75-6 ²⁹⁾³⁰⁾	Valve disc guide	✓	✓	✓	✓	-
2 per electric cable	81-39	Clamp	✓	✓	✓	✓	-
1	106	Suction casing	✓	-	-	✓	106 + 143, 412.01, 502, 900.01³²⁾, 901.02³³⁾, 920.02³³⁾
			-	✓	✓	-	106 + 143, 412.01, 502, 900.01³²⁾, 901.02³³⁾, 901.03, 920.02³³⁾, 920.03
1	108.01	Stage casing (last stage)	✓	✓	✓	✓	108.01 + 412.01, 529, 545, 902.01, 920.01
1 per stage -1	108.02	Stage casing	✓	✓	✓	✓	108.02 + 412.01, 502, 540, 902.01, 920.01
1	143 ²⁹⁾	Suction strainer	✓	✓	✓	✓	143 + 900.01³²⁾, 901.02³³⁾, 920.02³³⁾
1	211	Pump shaft	✓	-	-	-	211 + 412.02, 529, 540, 545, 550, 849, 901.04, 904.01, 940.01
			-	✓	-	-	211 + 412.02, 529, 540, 545, 550, 849, 901.04, 904.01, 904.02, 940.01, 940.02

29 Recommended spare part

30 Only for G6, NPT6-8 and DN 150 valve bodies

31 Only for G8, NPT8-8 and DN 200 valve bodies

32 Material variant C only

33 Material variant D only

Quantity	Part No.	Description	Motor size				Scope of supply of spare parts kit
			6 inch UMA 150, UMA-S 150	8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
1	211	Pump shaft	-	-	✓	✓	211 + 529, 540, 545, 550, 849, 901.04, 904.01, 940.01, 940.02
1 per stage	232	Clockwise impeller	✓	✓	✓	✓	232 + 52-4, 914
1 per stage +1	412.01 ²⁹⁾	O-ring	✓	✓	✓	✓	-
1	412.02 ²⁹⁾	O-ring	✓	✓	-	-	-
1	412.03 ²⁹⁾³¹⁾	O-ring	✓	✓	✓	✓	-
1 per stage	502 ²⁹⁾	Casing wear ring	✓	✓	✓	✓	-
1	529	Bearing sleeve	✓	✓	✓	✓	529 + 545
1 per stage -1	540 ²⁹⁾	Bush	✓	✓	✓	✓	-
1	545 ²⁹⁾	Bearing bush	✓	✓	✓	✓	545 + 529
1	550	Disc	✓	✓	✓	✓	-
1	751 ³⁰⁾	Valve body, flanged	✓	✓	✓	✓	751 + 75-6, 752.11, 752.12, 756, 759, 902.01, 920.01
		Valve body, threaded	✓	✓	✓	✓	751 + 75-6, 752.11, 752.12, 756, 759, 901.01, 902.01, 920.01
1	751 ³¹⁾	Valve body, flanged	✓	✓	✓	✓	751 + 752.12, 756, 759, 902.02, 920.04
		Valve body, threaded	✓	✓	✓	✓	751 + 752.12, 756, 759, 901.01, 902.02, 920.04
1	752.11 ³⁰⁾	Valve seat (metal)	✓	✓	✓	✓	752.11 + 752.12
1	752.12 ²⁹⁾	Valve seat (EPDM)	✓	✓	✓	✓	-
1	756 ²⁹⁾	Valve spring	✓	✓	✓	✓	-
1	759	Valve disc	✓	✓	✓	✓	759 + 752.11 ³⁰⁾ , 75-6 ³⁰⁾ , 752.12, 756
1 per electric cable	825	Cable guard	✓	✓	✓	✓	-
1	849	Sleeve coupling	✓	-	-	-	849 + 412.02, 904.01
			-	✓	-	-	849 + 412.02, 904.01, 904.02, 940.02
			-	-	✓	✓	849 + 904.01, 940.02
3	900.01 ³²⁾	Bolt/screw	✓	✓	✓	✓	-
4 per electric cable	900.02	Bolt/screw	✓	✓	✓	✓	-
2	901.01	Hexagon head bolt	✓	✓	✓	✓	-
3	901.02 ³³⁾	Hexagon head bolt	✓	✓	✓	✓	-
4	901.03	Hexagon head bolt	-	✓	✓	-	-
1	901.04	Hexagon head bolt	✓	✓	✓	✓	-
(1 per stage +1) × 12	902.01	Stud	✓	✓	✓	✓	-
12	902.02	Stud	✓	✓	✓	✓	-
1	904.01	Grub screw	✓	✓	-	-	-
4			-	-	✓	-	-
2			-	-	-	✓	-
2	904.02	Grub screw	-	✓	-	-	-
1 per stage × 4	914	Hexagon socket head cap screw	✓	✓	✓	✓	-
(1 per stage +1) × 12	920.01	Nut	✓	✓	✓	✓	-
3	920.02 ³³⁾	Nut	✓	✓	✓	✓	-
4	920.03	Nut	-	✓	✓	-	-
12	920.04	Nut	✓	✓	✓	✓	-
1	940.01 ²⁹⁾	Key	✓	✓	✓	✓	-
1	940.02 ²⁹⁾	Key	-	✓	✓	✓	-

Accessories

Installation parts

Table 42: Selection table

Description		Mat. No.	Weight [kg]
Set of pedestals for horizontal installation; comprising: 2 pedestals and fastening elements			
1.4301	UPA S 250 with UMA 150 or UMA-S 150 motor	01061868	
	UPA S 250 with UMA 200 or UMA-S 200 motor	01061869	
	UPA S 250 with UMA 250 or UMA-S 250 motor	01061870	
	UPA S 250 with UMA 300-250/2 motor with support frame	90068557	
1.4401	UPA S 250 with UMA 150 or UMA-S 150 motor	05100037	
	UPA S 250 with UMA 200 or UMA-S 200 motor	05100038	
	UPA S 250 with UMA 250 or UMA-S 250 motor	05100039	
	UPA S 250 with UMA 300-250/2 motor with support frame	05100040	
1.4539	UPA S 250 with UMA 150 or UMA-S 150 motor	05073108	
	UPA S 250 with UMA 200 or UMA-S 200 motor	05073109	
	UPA S 250 with UMA 250 or UMA-S 250 motor	05073110	
	UPA S 250 with UMA 300-250/2 motor with support frame	05073129	
 Bolts for fastening the pedestals to the foundation are not included in the scope of supply; they must be provided at the site.			
Suction shroud and pressure shroud		On request	-
Adapter		On request	-
Pair of supporting clamps and installation clamps for the following riser sizes, incl. fastening elements, material: painted steel			
R 6 / DN 150	L = 800 mm, F = 31,0 kN	95000304	29
R 8 / DN 200	L = 900 mm, F = 108 kN	95000307	70
 2 pairs are required for installation and dismantling.			

Centring device

Table 43: Centring device overview

	Motor size	Well diameter	Mat. No.			Weight
			Material variant			
		[mm]	Plastic	Galvanised steel	Stainless steel	[kg]
	UMA 150	250-299	90047662	-	-	-
	UMA-S 150	300-349	90047663	-	-	-
		350-400	90047664	-	-	-
		UMA 200	300-349	90047665	-	-
	UMA-S 200	350-400	90047666	-	-	-
	UMA-S 250	450-499	-	90037033	90037039	-
	UMA-S 250	500-549	-	90034400	90037041	-
		550-599	-	90037035	90037043	-
		600-650	-	90037037	90037045	-

Cooling jacket

Table 44: Cooling shroud overview

UPA S 250-...	Installation		Motor		Cooling jacket					Strainer				
	Ver- tical	Hori- zontal	Type	P _N	Ø × length	Weight	Material			Ø × length	Weight	Material		
				[kW]			1.4301/ AISI304SS	1.4401/ AISI316SS	1.4539/ AISI904L			1.4301/ AISI304SS	1.4401/ AISI316SS	1.4539/ AISI904L
					[mm]	[kg]				[mm]	[kg]			
UPA S 250 - 140	✓	-	UMA 150	18,0 - 22,0	Ø 285 (300) × 1250	14,4	01585004	01918300	01918456	Ø 285 × 385	1,9	01138989	01918565	01918577
			UMA 150	26,0-37,0	Ø 285 (300) × 1800	22,4	01585023	01918302	01918458	Ø 285 × 385	1,9	01138989	01918565	01918577
			UMA 200	45,0-65,0	Ø 285 (300) × 1800	20,9	01138800	01918304	01918460	Ø 285 × 385	1,9	01138989	01918565	01918577
			UMA 200	75,0-90,0	Ø 285 (300) × 2000	22,4	01138806	01918305	01918461	Ø 285 × 385	1,9	01138989	01918565	01918577
			UMA 250	85,0-110,0	Ø 330 (350) × 1800	26,0	01138824	01918314	01918470	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA 250	132,0	Ø 330 (350) × 2000	27,0	01138825	01918315	01918471	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA 250	160,0-190,0	Ø 330 (350) × 2250	29,0	01138826	01918316	01918472	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA S 150	18,5 - 37,0	Ø 285 (300) × 1250	14,4	01585004	01918300	01918456	Ø 285 × 385	1,9	01138989	01918565	01918577
			UMA S 200	75,0-130,0	Ø 285 (300) × 1800	20,9	01138800	01918304	01918460	Ø 285 × 385	1,9	01138989	01918565	01918577
			UMA S 250	200,0	Ø 330 (350) × 2000	27	01138825	01918315	01918471	Ø 330 × 385	2,3	01138990	01918566	01918578
	-	✓	UMA 150	18,0 - 22,0	Ø 285 (300) × 1250	22,4	01585021	01918359	01918515	Ø 285 × 385	1,9	01138989	01918565	01918577
			UMA 150	26,0-30,0	Ø 285 (300) × 1800	30,4	01585024	01918360	01918516	Ø 285 × 385	1,9	01138989	01918565	01918577
			UMA 200	45,0-65,0	Ø 285 (300) × 1800	28,9	01138939	01918362	01918518	Ø 285 × 385	1,9	01138989	01918565	01918577
			UMA 200	75,0	Ø 285 (300) × 2000	30,4	01138940	01918363	01918519	Ø 285 × 385	1,9	01138989	01918565	01918577
			UMA 250	85,0-110,0	Ø 330 (350) × 1800	35,0	01138947	01918370	01918526	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA 250	132,0	Ø 330 (350) × 2000	36,0	01138948	01918371	01918527	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA 250	160,0	Ø 330 (350) × 2250	38,0	01138949	01918372	01918528	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA S 150	18,5 - 37,0	Ø 285 (300) × 1250	22,4	01585021	01918359	01918515	Ø 285 × 385	1,9	01138989	01918565	01918577
			UMA S 200	75,0-130,0	Ø 285 (300) × 1800	20,9	01138800	01918304	01918460	Ø 285 × 385	1,9	01138989	01918565	01918577
			UMA S 250	200,0	Ø 330 (350) × 2000	36	01138948	01918371	01918527	Ø 330 × 385	2,3	01138990	01918566	01918578
UPA S 250-190 UPA S 250-240	✓	-	UMA 150	18,0 - 22,0	Ø 330 (350) × 1250	15,5	05190186	05190187	05190188	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA 150	26,0-30,0	Ø 330 (350) × 1500	17,9	01585026	01918308	01918464	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA 150	37,0	Ø 330 (350) × 1800	27,4	01585030	01918310	01918466	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA 200	37,0-45,0	Ø 330 (350) × 1500	23,9	01138821	01918311	01918467	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA 200	55,0-75,0	Ø 330 (350) × 1800	26,5	01138822	01918312	01918468	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA 200	90,0	Ø 330 (350) × 2000	28,2	01138823	01918313	01918469	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA 250	85,0-110,0	Ø 380 (400) × 1800	48,0	01138827	01918317	01918473	Ø 380 × 385	2,4	01138991	01918567	01918579
			UMA 250	132,0	Ø 380 (400) × 2000	51,0	01138829	01918318	01918474	Ø 380 × 385	2,4	01138991	01918567	01918579
			UMA 250	160,0	Ø 380 (400) × 2250	54,0	01138833	01918319	01918475	Ø 380 × 385	2,4	01138991	01918567	01918579
			UMA 300	250,0-300,0	Ø 420 (445) × 2500	68,0	05139619	-	05139942	Ø 420 × 385	4,0	01918580	-	01918580
			UMA S 150	18,5 - 37,0	Ø 330 (350) × 1250	15,5	05190186	05190187	05190188	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA S 200	75,0-130,0	Ø 330 (350) × 1800	26,5	01138822	01918312	01918468	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA S 250	200,0	Ø 380 (400) × 2000	51,0	01138829	01918318	01918474	Ø 380 × 385	2,4	01138991	01918567	01918579
	-	✓	UMA 150	18,0 - 22,0	Ø 330 (350) × 1250	22,0	05190220	05190221	05190222	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA 150	26,0-30,0	Ø 330 (350) × 1500	24,0	01585027	01918364	01918520	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA 200	37,0-45,0	Ø 330 (350) × 1500	32,9	01138944	01918367	01918523	Ø 330 × 385	2,3	01138990	01918566	01918578

UPA S 250-...	Installation		Motor		Cooling jacket					Strainer				
	Ver- tical	Hori- zontal			Type	P _N	Ø × length	Weight	Material			Ø × length	Weight	Material
			1.4301/ AISI304SS	1.4401/ AISI316SS					1.4539/ AISI904L	1.4301/ AISI304SS	1.4401/ AISI316SS			1.4539/ AISI904L
			[kW]	[mm]	[kg]	[mm]	[kg]							
UPA S 250-190	-	✓	UMA 200	55,0-75,0	Ø 330 (350) × 1800	35,5	01138945	01918368	01918524	Ø 330 × 385	2,3	01138990	01918566	01918578
UPA S 250-240			UMA 250	85,0-110,0	Ø 380 (400) × 1800	57,9	01138950	01918373	01918529	Ø 380 × 385	2,4	01138991	01918567	01918579
			UMA 250	132,0	Ø 380 (400) × 2000	60,9	01138951	01918374	01918530	Ø 380 × 385	2,4	01138991	01918567	01918579
			UMA 250	160,0	Ø 380 (400) × 2250	63,9	01138952	01918375	01918531	Ø 380 × 385	2,4	01138991	01918567	01918579
			UMA S 150	18,5 - 37,0	Ø 330 (350) × 1250	22,0	05190220	05190221	05190222	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA S 200	75,0-130,0	Ø 330 (350) × 1800	35,5	01138945	01918368	01918524	Ø 330 × 385	2,3	01138990	01918566	01918578
			UMA S 250	200,0	Ø 380 (400) × 2000	51	01138829	01918318	01918474	Ø 380 × 385	2,4	01138991	01918567	01918579

Flow velocity past the motor

Application

- Required motor cooling

Flow velocity past the motor

$v = 0 \text{ m/s}$

- The flow velocity past the motor is undefined.
- A free flow of heat is established. This heat flow must not be influenced or hindered by any structures, and a supply of fresh water must be provided at all times.
 - Example: pump set freely suspended in a vertical position inside a large tank

Flow velocity past the motor

$v > 0.2 \text{ m/s}$

$v > 0.5 \text{ m/s}$

- The flow velocity past the motor is defined for the installation conditions, see table below.
Decisive factors are the inside well diameter or the inside shroud diameter, the operating data and the outside dimensions of the pump set.
 - Example: pump set installed in a vertical position above the screen/filter in a well; values in accordance with the table below.
 - Example: pump set installed in a horizontal position in a tank, fitted with a cooling shroud, or pump set installed in a vertical position in a pump sump, fitted with a cooling shroud; values in accordance with the table below.

Table 45: Maximum permissible well diameter or inside shroud diameter

Flow rate	Flow velocity	Well diameter or inside shroud diameter [mm]				
Q [m³/h]	v [m/s]	UMA 150 UMA-S 150	UMA 200 UMA-S 200	UMA 250 UMA-S 250	UMA 300	14D
15	≥ 0,2	≤ 215	-	-	-	-
	≥ 0,5	≤ 175	-	-	-	-
25	≥ 0,2	≤ 255	-	-	-	-
	≥ 0,5	≤ 195	-	-	-	-
50	≥ 0,2	≤ 330	≤ 350	-	-	-
	≥ 0,5	≤ 235	≤ 265	-	-	-
75	≥ 0,2	≤ 390	≤ 410	≤ 430	-	-
	≥ 0,5	≤ 270	≤ 300	≤ 330	-	-
100	≥ 0,2	≤ 445	≤ 460	≤ 480	-	-
	≥ 0,5	≤ 300	≤ 325	≤ 355	-	-
125	≥ 0,2	≤ 490	≤ 510	≤ 525	-	-
	≥ 0,5	≤ 330	≤ 350	≤ 380	-	-
150	≥ 0,2	≤ 535	≤ 550	≤ 565	≤ 590	-
	≥ 0,5	≤ 355	≤ 380	≤ 400	≤ 430	-
175	≥ 0,2	≤ 575	≤ 590	≤ 605	≤ 625	-
	≥ 0,5	≤ 380	≤ 400	≤ 420	≤ 450	-
200	≥ 0,2	≤ 615	≤ 625	≤ 640	≤ 660	≤ 690
	≥ 0,5	≤ 405	≤ 420	≤ 445	≤ 470	≤ 510
250	≥ 0,2	≤ 680	≤ 690	≤ 705	≤ 725	≤ 750
	≥ 0,5	≤ 445	≤ 460	≤ 480	≤ 505	≤ 540
300	≥ 0,2	≤ 745	≤ 755	≤ 765	≤ 780	≤ 800
	≥ 0,5	≤ 485	≤ 500	≤ 515	≤ 540	≤ 570
350	≥ 0,2	-	≤ 810	≤ 820	≤ 835	≤ 860
	≥ 0,5	-	≤ 530	≤ 550	≤ 570	≤ 600
400	≥ 0,2	-	≤ 865	≤ 875	≤ 890	≤ 910
	≥ 0,5	-	≤ 565	≤ 580	≤ 605	≤ 630
500	≥ 0,2	-	≤ 960	≤ 970	≤ 985	≤ 1000
	≥ 0,5	-	≤ 625	≤ 640	≤ 660	≤ 690
600	≥ 0,2	-	≤ 1050	≤ 1055	≤ 1070	≤ 1090
	≥ 0,5	-	≤ 680	≤ 695	≤ 710	≤ 740
800	≥ 0,2	-	≤ 1205	≤ 1215	≤ 1225	≤ 1240
	≥ 0,5	-	≤ 775	≤ 790	≤ 805	≤ 830
1000	≥ 0,2	-	≤ 1345	≤ 1350	≤ 1360	≤ 1370
	≥ 0,5	-	≤ 865	≤ 875	≤ 890	≤ 910

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Flow rate	Flow velocity	Well diameter or inside shroud diameter [mm]				
Q [m³/h]	v [m/s]	UMA 150 UMA-S 150	UMA 200 UMA-S 200	UMA 250 UMA-S 250	UMA 300	14D
1200	≥ 0,2	-	-	-	≤ 1485	≤ 1500
	≥ 0,5	-	-	-	≤ 965	≤ 980
1400	≥ 0,2	-	-	-	≤ 1600	≤ 1610
	≥ 0,5	-	-	-	≤ 1030	≤ 1050
1600	≥ 0,2	-	-	-	≤ 1705	≤ 1720
	≥ 0,5	-	-	-	≤ 1100	≤ 1120
1800	≥ 0,2	-	-	-	≤ 1805	≤ 1820
	≥ 0,5	-	-	-	≤ 1165	≤ 1180
2000	≥ 0,2	-	-	-	≤ 1900	≤ 1910
	≥ 0,5	-	-	-	≤ 1225	≤ 1240

Power cables

Main applications

- For use with drinking water applications
- Motor lead for submersible motors, laid under water
- Extension cable for the motor lead, laid freely exposed to air and in contact with surfaces
- Unshielded and shielded cable versions

Operating data

Table 46: Operating properties

Characteristic		Value
Rated voltage	V_N [V]	≤ 1000
Ambient temperature	T [°C]	≤ 50
Immersion depth	IM [m]	≤ 500

Designation

Unshielded cable

Example: ZN 1391 - G FL GWT - J 4G25 - Cu-rubber

Table 47: Designation key for unshielded cable

Abbreviation	Description
ZN	Company standard
G	Insulation, rubber
FL	Type of power cable
	FL Flat
	RD Round
GWT	Suitable for drinking water
J	Information about earth conductor
	J With earth conductor
	O Without earth conductor
4G25	Number of cores
	4G25 With earth conductor, 4-core, cross-section 25 mm ²
	3 × 25 Without earth conductor, 3-core, cross-section 25 mm ²
Cu-rubber	Material

Shielded cable

Example: ZN 1391 - S 07 B C4 B - F 3X6 KON

Table 48: Designation key for shielded cable

Abbreviation	Description
ZN	Company standard
S	Basic type according to manufacturer
07	Rated voltage U_0/U
	05 300/500 V
	07 450/750 V
	1/10 600/1000 V
B	Insulation material
	B EPR
C4	Braided shield
B	Sheath material
	B EPR
F	Flexible

Abbreviation	Description
3X6	Nominal cross-section per phase
	3 × 6 Without earth conductor, 3-core, cross-section 6 mm ²
	4G35 With earth conductor, 4-core, cross-section 35 mm ²
KON	Earth conductor design
	KON Earth conductor, concentric as braided shield
	3G16/3 Earth conductor cross-section = nominal phase cross-section (earth conductor divided into three cores, earth conductor cross-section = 16 mm ²), G = with earth conductor

Design details

Unshielded rubber-sheathed cable

Comprising:

- Conductor
 - Copper
 - Bare
 - Fine-stranded
 - Class 5 to DIN EN 60228
- Insulation
 - EPR-based insulation compound (ethylene-propylene rubber) resistant to ozone, weather and water
- Core identification
 - Core colours to DIN EN 50525-1
- Sheath
 - Core insulation made of EPR
 - Outer sheath made of EPDM rubber compound
 - Colour: blue

Shielded rubber-sheathed cable

Comprising:

- Conductor
 - Copper
 - Bare
 - Fine-stranded
 - Class 5 to DIN VDE 0295 / IEC 60228
- Insulation
 - EPR-based insulation compound (ethylene-propylene rubber) resistant to ozone, weather and water
- Core identification
 - Core colours to DIN VDE 0293-308:2003
- Two-layer sheath design
 - Inner sheath: Special EPR compound as per German KTW recommendations, colour: blue
 - Outer sheath: Special EPR compound as per German KTW recommendations, DVGW (Worksheet 270), WRAS and ACS, colour: blue
- Shielding
 - Tinned copper wire braiding over the inner layer of the sheath system, max. coupling resistance 30 Ohm/km at 30 MHz
- Marking
 - HYDROFIRM(T) EMV-FC S1BC4B-F 3X10/10 KON 0.6/1 kV KTW DVGW W270 ACS WRAS

Table 49: Overview of electric cables

Type of electric cable	Characteristic
Flat cable, unshielded	
	3 cores, flat
	4 cores, flat

Type of electric cable	Characteristic
Round cable, unshielded	
	1 core, round
	4 cores, round
Round cable, shielded	
	4 cores, round

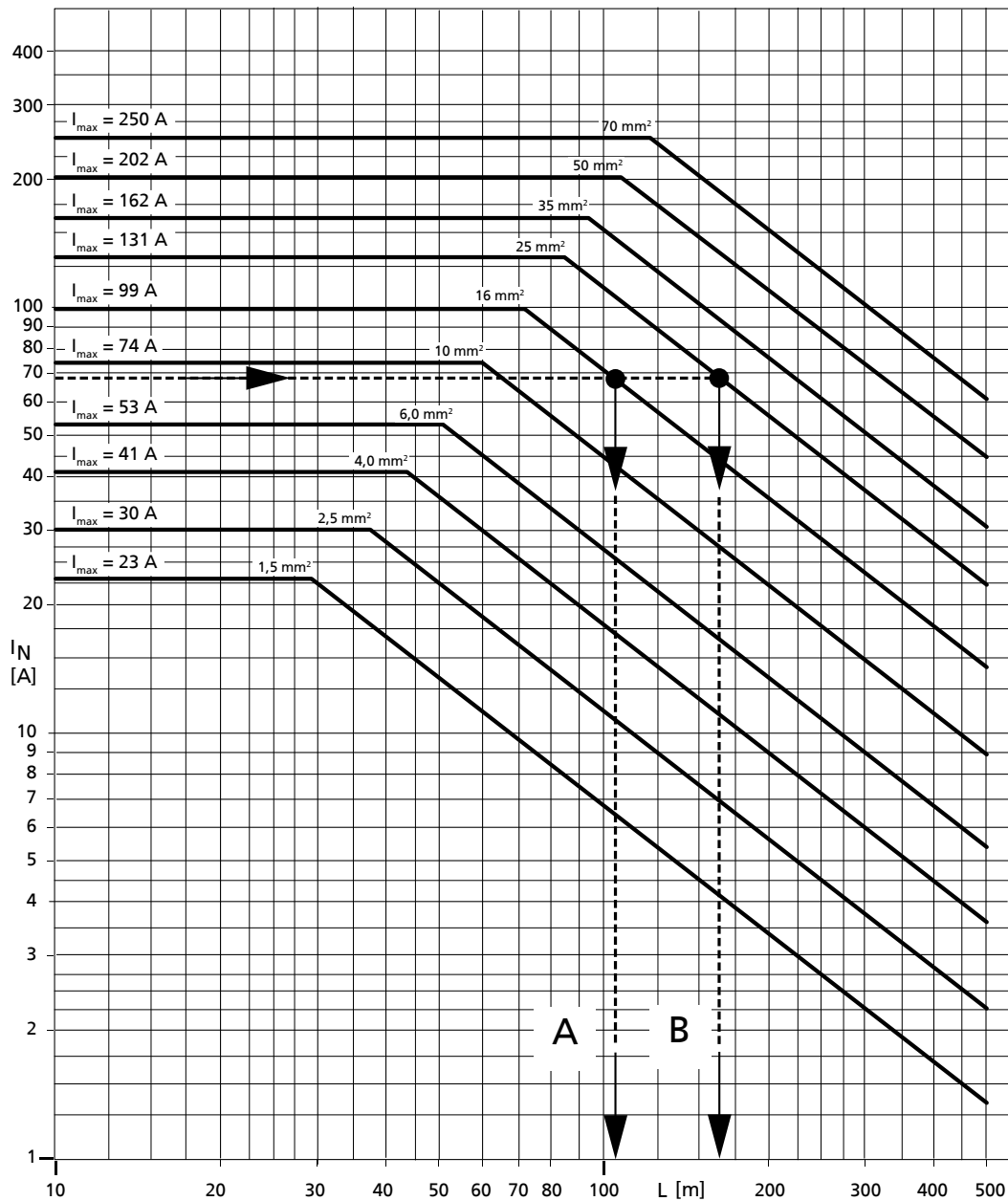
Selection information

Permissible cable lengths for starting method: DOL

Applies to 1 cable or 2 cables in parallel³⁴⁾

Conditions:

- $V = 400 \text{ V}$; $\Delta V = 3 \%$; $T \leq 30 \text{ °C}$; laid freely exposed to air and in contact with surfaces



Example A

$I_N = 68 \text{ A}$
Cable design: 1 x 16 mm^2
Cable length: $L \leq 105 \text{ m}$

Example B

$I_N = 68 \text{ A}$
Cable design: 1 x 25 mm^2
Cable length: $L \leq 165 \text{ m}$

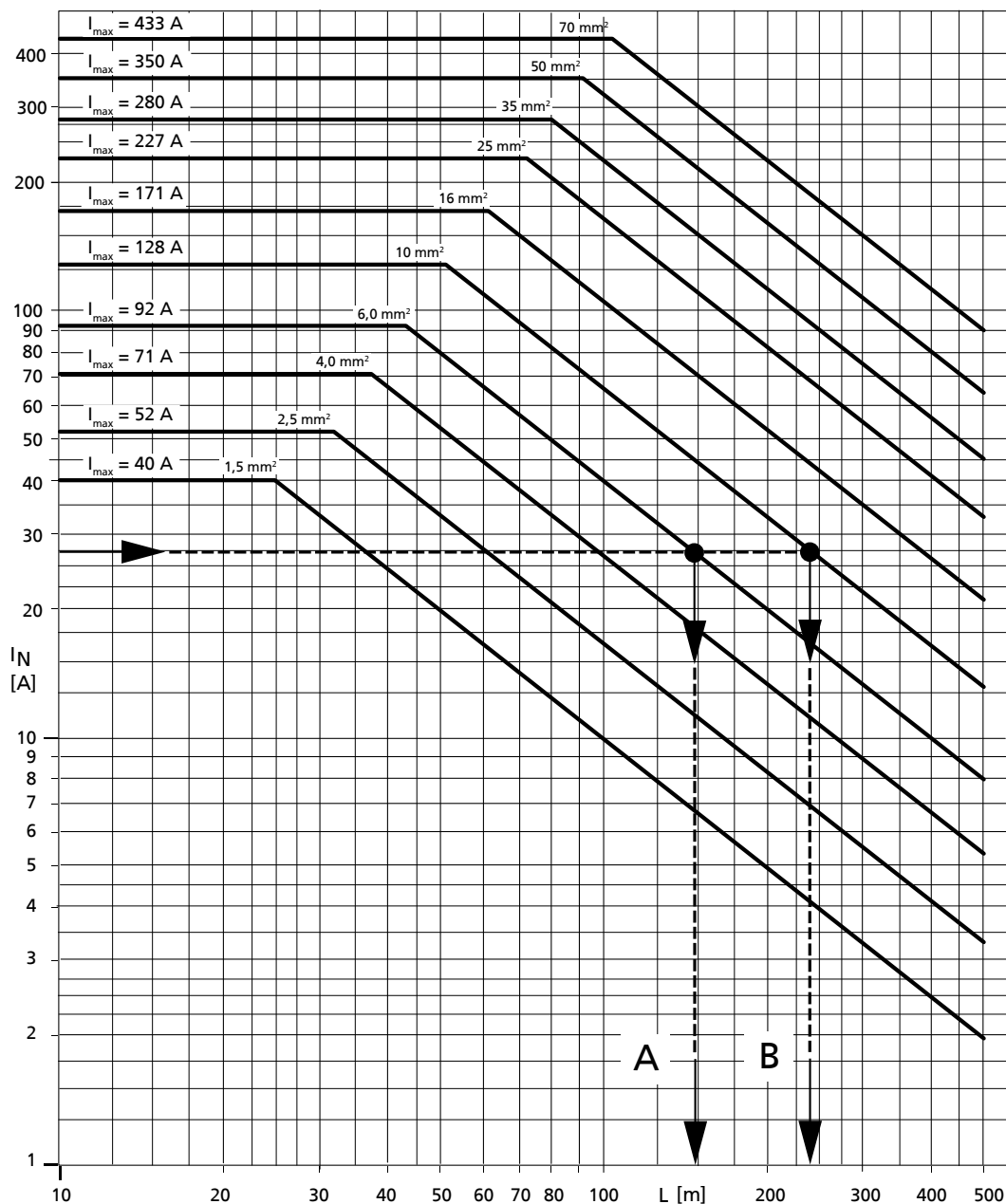
³⁴ For parallel cables the value is doubled to obtain the permissible length.

Permissible cable lengths for starting method: YΔ

Applies to 2 cables

Conditions:

- $V = 400 \text{ V}$; $\Delta V = 3 \%$; $T \leq 30 \text{ °C}$; laid freely exposed to air and in contact with surfaces



Example A

$I_N = 27 \text{ A}$
Cable design: $2 \times 6.0 \text{ mm}^2$
Cable length: $L \leq 145 \text{ m}$

Example B

$I_N = 27 \text{ A}$
Cable design: $2 \times 10 \text{ mm}^2$
Cable length: $L \leq 235 \text{ m}$

Voltage drop in the extension cable

When calculating the required conductor cross-section q not only I_N , but also the voltage drop ΔV along the cable length L (distance from motor to control unit) must be taken into account. Proper functioning of our submersible motors requires $\Delta V \leq 3 \%$ of the

supply voltage V .

If $\Delta V > 3 \%$, a larger conductor cross-section must be used. The voltage drop is calculated using the following equations:

Starting method DOL/autotransformer

- 1 cable:

$$\Delta U = \frac{3,1 \times L \times I_N \times \cos \varphi}{q \times U} [\%]$$

- 2 cables in parallel (II):

$$\Delta U = \frac{1,55 \times L \times I_N \times \cos \varphi}{q \times U} [\%]$$

Starting method YΔ (2 cables):

$$\Delta U = \frac{2,1 \times L \times I_N \times \cos \varphi}{q \times U} [\%]$$

Power loss ΔP:

$$\Delta P = \frac{\Delta U}{(\cos \varphi)^2} [\%]$$

Table 50: Key

Code	Description
L	Single cable length [m]
I _N	Rated current [A]:
cos φ	Power factor at 4/4 load
q	Conductor cross-section [mm ²]
U	Supply voltage [V]

Maximum permissible rated motor current
Table 51: For ambient temperature t ≤ 30 °C

Starting method	Used as ...	I _{max} [A] for the following conductor cross-sections [mm ²]										
		1,5	2,5	4	6	10	16	25	35	50	70	95
DOL (1 cable or 2 cables in parallel)	Motor lead	29	38	52	67	94	125	166	205	256	316	517
	Extension cable	23	30	41	53	74	99	131	162	202	250	409
YΔ (2 cables)	Motor lead	50	66	90	116	163	217	288	355	443	547	895
	Extension cable	40	52	71	92	128	171	227	280	350	433	708

Dimensions and weights
Table 52: Technical data, unshielded electric extension cable for connection with the motor lead

Core type	Description	Cable cross-section	Height	Width	Diameter	Cable designation	Dimensions	Mat. No.	Weight
		[mm ²]	[mm]	[mm]	[mm]				[kg/m]
●●●	Unshielded electric extension cable, 3-core, flat, without earth conductor	1,5	5,2 ^{+1,0}	11,0 ^{+2,0}	-	G FL GWT-O	3×1,5	90068148	0.11
		2,5	6,1 ^{+1,5}	13,2 ^{+2,3}	-	G FL GWT-O	3×2,5	90068149	0.17
		4	7,0 ^{+2,0}	15,5 ^{+3,5}	-	G FL GWT-O	3×4	90068150	0.25
		6	7,6 ^{+2,2}	17,4 ^{+3,6}	-	G FL GWT-O	3×6	90068151	0.32
		10	9,3 ^{+2,2}	21,5 ^{+3,5}	-	G FL GWT-O	3×10	90068152	0.49
		16	11,2 ^{+2,3}	26,7 ^{+4,3}	-	G FL GWT-O	3×16	90068153	0.75
		25	13,0 ^{+2,5}	31,6 ^{+3,9}	-	G FL GWT-O	3×25	90068154	1.11
		35	14,6 ^{+2,9}	35,5 ^{+5,0}	-	G FL GWT-O	3×35	90068155	1.44
		50	17,0 ^{+3,0}	42,1 ^{+4,9}	-	G FL GWT-O	3×50	90068156	2.05
		70	19,3 ^{+2,7}	48,4 ^{+3,6}	-	G FL GWT-O	3×70	90068157	2.76
		95	-	-	-	-	-	-	-
●●●●	Unshielded electric extension cable, 4-core, flat, with earth conductor	1,5	5,2 ^{+1,0}	14,5 ^{+2,7}	-	G FL GWT-J	4G×1,5	90068158	0.17
		2,5	6,1 ^{+1,5}	17,5 ^{+2,5}	-	G FL GWT-J	4G×2,5	90068159	0.24
		4	-	-	-	G FL GWT-J	4G×4	90068160	0.34
		6	7,6 ^{+2,2}	23,5 ^{+3,0}	-	G FL GWT-J	4G×6	90068161	0.44
		10	9,3 ^{+2,2}	29,0 ^{+3,5}	-	G FL GWT-J	4G×10	90068162	0.7
		16	11,2 ^{+2,3}	35,0 ^{+2,4}	-	G FL GWT-J	4G×16	90068163	1.03
		25	13,0 ^{+2,5}	41,5 ^{+4,5}	-	G FL GWT-J	4G×25	90068164	1.46
		35	-	-	-	G FL GWT-J	4G×25	90068165	1.96
		50	-	-	-	-	-	-	-
		70	-	-	-	-	-	-	-
		95	-	-	-	-	-	-	-
●	Unshielded electric extension cable, 1-core, round, without earth conductor	1,5	-	-	5,3 ^{+1,1}	G RD GWT-O	1×1,5	01046306	0.01
		2,5	-	-	-	-	-	-	-
		4	-	-	-	-	-	-	-
		6	-	-	7,4 ^{+1,2}	G RD GWT-O	1×6	01042783	0.11

Core type	Description	Cable cross-section	Height	Width	Diameter	Cable designation	Dimensions	Mat. No.	Weight
		[mm ²]	[mm]	[mm]	[mm]				[kg/m]
	Unshielded electric extension cable, 1-core, round, without earth conductor	10	-	-	9,1 ^{+3,4}	G RD GWT-O	1x10	01042784	0.18
		16	-	-	10,7 ^{+2,4}	G RD GWT-O	1x16	01042785	0.27
		25	-	-	12,6 ^{+2,6}	G RD GWT-O	1x25	01042786	0.38
		35	-	-	13,8 ^{+3,6}	G RD GWT-O	1x35	01042787	0.5
		50	-	-	16,2 ^{+3,6}	G RD GWT-O	1x50	01042788	0.7
		70	-	-	18,5 ^{+3,6}	G RD GWT-O	1x70	01042789	0.94
		95	-	-	21,9 ^{+1,5}	G RD GWT-O	1x95	01042794	1.15
	Unshielded electric extension cable, 4-core, round, with earth conductor	1,5	-	-	10,0 ^{+2,0}	G RD GWT-J	4G1,5	90068174	0.18
		2,5	-	-	12,0 ^{+1,9}	G RD GWT-J	4G2,5	90068175	0.26
		4	-	-	13,9 ^{+2,0}	G RD GWT-J	4G4	90068176	0.36
		6	-	-	15,7 ^{+2,1}	G RD GWT-J	4G6	90068177	0.48
		10	-	-	21,1 ^{+2,1}	G RD GWT-J	4G10	90068178	0.84
		16	-	-	24,5 ^{+4,3}	G RD GWT-J	4G16	90068179	1.22
		25	-	-	29,7 ^{+4,3}	G RD GWT-J	4G25	90068180	1.77
		35	-	-	33,3 ^{+5,5}	G RD GWT-J	4G35	90068181	2.3
		50	-	-	39,0 ^{+5,6}	G RD GWT-J	4G50	90068182	3.19
		70	-	-	44,2 ^{+5,8}	G RD GWT-J	4G70	90068183	4.36
		95	-	-	-	-	-	-	-

Table 53: Technical data, shielded electric extension cable for connection with the motor lead

Core type	Description	Cable cross-section		Cable designation	Dimensions	Mat. No.	Weight
		[mm ²]	Diameter [mm]				[kg/m]
	Shielded electric extension cable, 4-core, round	0,5	7,5 ^{+1,0}	S 05 B C4 B-F	F4x0,5	05536014	0.09
		6	17,8 ^{+2,0}	S 07 B C4 B-F	3x6/6 KON	05535908	0.5
		16	24,7 ^{+3,0}	S 07 B C4 B-F	3x16+3G16/3	05535909	1.25
		35	32,1 ^{+2,0}	S 07 B C4 B-F	3x35+3G16/3	05536010	2.1
		50	36,8 ^{+2,0}	S 07 B C4 B-F	3x50+3G25/3	05536011	2.78
		70	40,4 ^{+3,0}	S 07 B C4 B-F	3x70+3G35/3	05536012	3.77
		95	45,0 ^{+2,0}	S 07 B C4 B-F	3x95+3G50/3	05536013	4.85

Cable connector

Application

For providing a sealed, water-tight connection between electric extension cables and motor leads of submersible motors.

- Ambient temperature: $T \leq +50\text{ °C}$
- Immersion depth: $\leq 500\text{ m}$

Selection table

The table below only shows the **electric cables on stock**. Cable connectors for other electric cables on request

Table 54: Key

Cable symbols	Explanation	Cable designation
	1-core, round, unshielded	G RD GWT - O 1 G
	4-core, round, unshielded	G RD GWT - J 4 G
	3-core, flat, unshielded	G FL GWT - O 3x ...
	4-core, flat, unshielded	G FL GWT - J 4G ...
	4-core, round, shielded	S 07 B C4 B-F - 4G35

Cable connector for an electric extension cable

Table 55: Cable connector: electric extension cable, →

Motor lead [mm²]	Electric extension cable [mm²]										
	1,5	2,5	4	6	10	16	25	35	50	70	95
1,5	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	-	-	-
2,5	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	-	-	-
4	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	-	-	-
6	-	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	-	-	-
	-	-	-	-	-	29F 95005160 ⁽³⁵⁾	29F 95005160 ⁽³⁵⁾	39F 95005510 ⁽³⁵⁾	-	-	-
10	-	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90053910 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90053910 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90053910 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	-	-
	-	-	-	-	29F 95005159 ⁽³⁵⁾	29F 95005161 ⁽³⁵⁾	29F 95005161 ⁽³⁵⁾	39F 95005176 ⁽³⁵⁾	-	-	-
16	-	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90053910 ⁽³⁶⁾	53 95005192 ⁽³⁵⁾ 90053913 ⁽³⁶⁾	53 95005192 ⁽³⁵⁾ 90053913 ⁽³⁶⁾	53 95005192 ⁽³⁵⁾ 90053913 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	-
25	-	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 95005192 ⁽³⁵⁾ 90053913 ⁽³⁶⁾	66 95005214 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005214 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	-
35	-	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 95005192 ⁽³⁵⁾ 90053913 ⁽³⁶⁾	66 95005214 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005206 ⁽³⁵⁾ 90053907 ⁽³⁶⁾	66 95005206 ⁽³⁵⁾ 90053907 ⁽³⁶⁾	66 95005206 ⁽³⁵⁾ 90053907 ⁽³⁶⁾	-
50	-	-	-	-	-	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005206 ⁽³⁵⁾ 90053907 ⁽³⁶⁾	66 95005206 ⁽³⁵⁾ 90053907 ⁽³⁶⁾	66 95005206 ⁽³⁵⁾ 90053907 ⁽³⁶⁾	-
70	-	-	-	-	-	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005206 ⁽³⁵⁾ 90053907 ⁽³⁶⁾	66 95005206 ⁽³⁵⁾ 90053907 ⁽³⁶⁾	66 95005207 ⁽³⁵⁾ 90053907 ⁽³⁶⁾	-

35 MVB - material No. for cable connector, extended and sealed during manufacture

36 OVB - material No. for cable connector, not connected and supplied but not fitted

Table 56: Cable connector: electric extension cable, →

Motor lead [mm²]	Electric extension cable [mm²]										
	1,5	2,5	4	6	10	16	25	35	50	70	95
1,5	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	-	-	-
2,5	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	-	-	-
4	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	-	-	-
6	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	-	-	-
10	43 90049389 ⁽³⁵⁾ 90053910 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90053910 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90053910 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90053910 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90053910 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90053910 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	-	-
16	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90053910 ⁽³⁶⁾	53 95005192 ⁽³⁵⁾ 90053913 ⁽³⁶⁾	53 95005192 ⁽³⁵⁾ 90053913 ⁽³⁶⁾	53 95005192 ⁽³⁵⁾ 90053913 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	-
25	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 95005192 ⁽³⁵⁾ 90053913 ⁽³⁶⁾	66 95005214 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005214 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	-

Table 57: Cable connector: electric extension cable, + or + →

Motor lead [mm²]	Electric extension cable [mm²]										
	1,5	2,5	4	6	10	16	25	35	50	70	95
+ or +											
1,5	28 90049386 ⁽³⁵⁾ 90049396 ⁽³⁶⁾	28 90049386 ⁽³⁵⁾ 90049396 ⁽³⁶⁾	28 90049386 ⁽³⁵⁾ 90049396 ⁽³⁶⁾	35 95005165 ⁽³⁵⁾ 90039349 ⁽³⁶⁾	35 95005165 ⁽³⁵⁾ 90039349 ⁽³⁶⁾	-	-	-	-	-	-
2,5	28 90049388 ⁽³⁵⁾ 90049398 ⁽³⁶⁾	28 90049388 ⁽³⁵⁾ 90049398 ⁽³⁶⁾	28 90049388 ⁽³⁵⁾ 90049398 ⁽³⁶⁾	35 95005165 ⁽³⁵⁾ 90039349 ⁽³⁶⁾	35 95005165 ⁽³⁵⁾ 90039349 ⁽³⁶⁾	-	-	-	-	-	-
4	-	-	35 95005165 ⁽³⁵⁾ 90039349 ⁽³⁶⁾	35 95005165 ⁽³⁵⁾ 90039349 ⁽³⁶⁾	35 95005165 ⁽³⁵⁾ 90039349 ⁽³⁶⁾	43 90049390 ⁽³⁵⁾ 90049400 ⁽³⁶⁾	-	-	-	-	-
6	-	-	-	43 90049390 ⁽³⁵⁾ 90049400 ⁽³⁶⁾	43 90049390 ⁽³⁵⁾ 90049400 ⁽³⁶⁾	43 90049390 ⁽³⁵⁾ 90049400 ⁽³⁶⁾	53 90049392 ⁽³⁵⁾ 90049402 ⁽³⁶⁾	53 90049392 ⁽³⁵⁾ 90049402 ⁽³⁶⁾	66 90033248 ⁽³⁵⁾ 90055737 ⁽³⁶⁾	66 90033248 ⁽³⁵⁾ 90055737 ⁽³⁶⁾	-
10	-	-	-	66 95005522 ⁽³⁵⁾ 90054294 ⁽³⁶⁾	66 95005522 ⁽³⁵⁾ 90054294 ⁽³⁶⁾	66 90039412 ⁽³⁵⁾ 90053905 ⁽³⁶⁾	66 90039412 ⁽³⁵⁾ 90053905 ⁽³⁶⁾	66 90039412 ⁽³⁵⁾ 90053905 ⁽³⁶⁾	66 90039412 ⁽³⁵⁾ 90053905 ⁽³⁶⁾	66 90033248 ⁽³⁵⁾ 90055737 ⁽³⁶⁾	-
16	-	-	-	-	-	66 90039361 ⁽³⁵⁾ 90053906 ⁽³⁶⁾	66 90039361 ⁽³⁵⁾ 90053906 ⁽³⁶⁾	66 90039361 ⁽³⁵⁾ 90053906 ⁽³⁶⁾	66 90036538 ⁽³⁵⁾ 90055738 ⁽³⁶⁾	66 90036538 ⁽³⁵⁾ 90055738 ⁽³⁶⁾	-
25	-	-	-	-	-	-	78 90039701 ⁽³⁵⁾ 90053908 ⁽³⁶⁾	78 90039701 ⁽³⁵⁾ 90053908 ⁽³⁶⁾	78 90039701 ⁽³⁵⁾ 90053908 ⁽³⁶⁾	78 90039701 ⁽³⁵⁾ 90053908 ⁽³⁶⁾	-

Table 58: Cable connector: electric extension cable, →

Motor lead	Electric extension cable [mm ²]										
[mm ²]	1,5	2,5	4	6	10	16	25	35	50	70	95
1,5	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	-	-	-	-	-
2,5	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	-	-	-	-	-
4	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	-	-	-	-	-
6	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	28 90049385 ⁽³⁵⁾ 95005106 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	-	-	-
10	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	35 90049387 ⁽³⁵⁾ 90049397 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	66 95005523 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	-	-
16	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	43 90049389 ⁽³⁵⁾ 90049399 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	66 95005370 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005214 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	-
25	-	-	-	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	66 95005520 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	78 95005527 ⁽³⁵⁾	-
35	-	-	-	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	53 90049391 ⁽³⁵⁾ 90049401 ⁽³⁶⁾	66 95005214 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	78 95005527 ⁽³⁵⁾	-
50	-	-	-	-	-	66 95005370 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005520 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005214 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	66 95005214 ⁽³⁵⁾ 90053914 ⁽³⁶⁾	78 95005527 ⁽³⁵⁾	-
70	-	-	-	-	-	-	-	-	78 95005527 ⁽³⁵⁾	78 95005527 ⁽³⁵⁾	-

Table 59: Cable connector: extension cable for Pt100 sensor, →

Sensor cable for Pt100	Extension cable [mm ²]										
[mm ²]	0,5	-	-	-	-	-	-	-	-	-	-
0,5	18 90062234 ⁽³⁵⁾ 90062233 ⁽³⁶⁾	-	-	-	-	-	-	-	-	-	-

Table 60: Cable connector: extension cable for Pt100 sensor, →

Sensor cable for Pt100 [mm²]	Extension cable [mm²]										
	1,5	2,5	4	6	10	16	25	35	50	70	95
35	-	-	-	-	-	-	-	35 95005172 ³⁵⁾ 90023666 ³⁶⁾	-	-	-
50	-	-	-	-	-	-	-	-	35 95005173 ³⁵⁾ 90023666 ³⁶⁾	-	-
70	-	-	-	-	-	-	-	-	-	35 95005174 ³⁵⁾ 90023666 ³⁶⁾	-
95	-	-	-	-	-	-	-	-	-	-	43 95005188 ³⁵⁾

Cable connector for a shielded electric extension cable
Table 61: Cable connector: shielded electric extension cable, →

Motor lead [mm²]	Shielded electric extension cable [mm²]									
	1,5	2,5	4	6	10	16	25	35	50	70
1,5	-	-	-	-	-	-	-	-	-	-
2,5	-	-	-	35 90049387 ³⁷⁾ 90049397 ³⁸⁾	-	43 90049389 ³⁷⁾ 90049399 ³⁸⁾	-	53 90049391 ³⁷⁾ 90049401 ³⁸⁾	-	-
4	-	-	-	35 90049387 ³⁷⁾ 90049397 ³⁸⁾	-	43 90049389 ³⁷⁾ 90049399 ³⁸⁾	-	53 90049391 ³⁷⁾ 90049401 ³⁸⁾	-	-
6	-	-	-	35 90049387 ³⁷⁾ 90049397 ³⁸⁾	-	43 90049389 ³⁷⁾ 90049399 ³⁸⁾	-	53 90049391 ³⁷⁾ 90049401 ³⁸⁾	-	-
10	-	-	-	-	-	43 90049389 ³⁷⁾ 90053910 ³⁸⁾	-	53 90049391 ³⁷⁾ 90049401 ³⁸⁾	66 95005523 ³⁷⁾ 90053914 ³⁸⁾	-
16	-	-	-	-	-	53 95005192 ³⁷⁾ 90053913 ³⁸⁾	-	53 95005192 ³⁷⁾ 90053913 ³⁸⁾	66 95005523 ³⁷⁾ 90053914 ³⁸⁾	66 95005523 ³⁷⁾ 90053914 ³⁸⁾
25	-	-	-	-	-	-	-	66 95005214 ³⁷⁾ 90053914 ³⁸⁾	66 95005523 ³⁷⁾ 90053914 ³⁸⁾	66 95005523 ³⁷⁾ 90053914 ³⁸⁾

37 MVB - material No. for cable connector, extended and sealed during manufacture

38 OVB - material No. for cable connector, not connected and supplied but not fitted

Table 62: Cable connector: shielded electric extension cable, + or + →

Motor lead [mm²]	Shielded electric extension cable [mm²]									
	1,5	2,5	4	6	10	16	25	35	50	70
+ or +										
1,5	-	-	-	-	-	-	-	-	-	-
2,5	-	-	-	35 95005165 ³⁷⁾ 90039349 ³⁸⁾	-	-	-	-	-	-
4	-	-	-	-	-	43 90049390 ³⁷⁾ 90049400 ³⁸⁾	-	-	-	-
6	-	-	-	-	-	43 90049390 ³⁷⁾ 90049400 ³⁸⁾	-	53 90049392 ³⁷⁾ 90049402 ³⁸⁾	66 90033248 ³⁷⁾ 90055737 ³⁸⁾	66 90033248 ³⁷⁾ 90055737 ³⁸⁾
10	-	-	-	-	-	-	-	66 90039412 ³⁷⁾ 90053905 ³⁸⁾	66 90039412 ³⁷⁾ 90053905 ³⁸⁾	66 90033248 ³⁷⁾ 90055737 ³⁸⁾
16	-	-	-	-	-	-	-	66 90039361 ³⁷⁾ 90053906 ³⁸⁾	66 90036538 ³⁷⁾ 90055738 ³⁸⁾	66 90036538 ³⁷⁾ 90055738 ³⁸⁾
25	-	-	-	-	-	-	-	-	78 90039701 ³⁷⁾ 90053908 ³⁸⁾	78 90039701 ³⁷⁾ 90053908 ³⁸⁾
35	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-

Table 63: Cable connector: shielded electric extension cable, →

Motor lead [mm²]	Shielded electric extension cable [mm²]									
	1,5	2,5	4	6	10	16	25	35	50	70
1,5	-	-	-	-	-	-	-	-	-	-
2,5	-	-	-	28 90049385 ³⁷⁾ 95005106 ³⁸⁾	-	43 90049389 ³⁷⁾ 90049399 ³⁸⁾	-	-	-	-
4	-	-	-	28 90049385 ³⁷⁾ 95005106 ³⁸⁾	-	43 90049389 ³⁷⁾ 90049399 ³⁸⁾	-	-	-	-
6	-	-	-	28 90049385 ³⁷⁾ 95005106 ³⁸⁾	-	43 90049389 ³⁷⁾ 90049399 ³⁸⁾	-	53 90049391 ³⁷⁾ 90049401 ³⁸⁾	-	-
10	-	-	-	-	-	43 90049389 ³⁷⁾ 90049399 ³⁸⁾	-	53 90049391 ³⁷⁾ 90049401 ³⁸⁾	66 95005523 ³⁷⁾ 90053914 ³⁸⁾	-
16	-	-	-	-	-	43 90049389 ³⁷⁾ 90049399 ³⁸⁾	-	53 90049391 ³⁷⁾ 90049401 ³⁸⁾	66 95005370 ³⁷⁾ 90053914 ³⁸⁾	66 95005214 ³⁷⁾ 90053914 ³⁸⁾
25	-	-	-	-	-	-	-	53 90049391 ³⁷⁾ 90049401 ³⁸⁾	66 95005520 ³⁷⁾ 90053914 ³⁸⁾	78 95005527 ³⁷⁾ 05244394 ³⁸⁾

Motor lead	Shielded electric extension cable [mm ²]									
[mm ²]	1,5	2,5	4	6	10	16	25	35	50	70
										
35	-	-	-	-	-	-	-	53 90049391 ⁽³⁷⁾ 90049401 ⁽³⁸⁾	66 95005214 ⁽³⁷⁾ 90053914 ⁽³⁸⁾	78 95005527 ⁽³⁷⁾ 05244395 ⁽³⁸⁾
50	-	-	-	-	-	-	-	-	66 95005214 ⁽³⁷⁾ 90053914 ⁽³⁸⁾	78 95005527 ⁽³⁷⁾ 05244396 ⁽³⁸⁾
70	-	-	-	-	-	-	-	-	78 95005527 ⁽³⁷⁾ 05244396 ⁽³⁸⁾	78 95005527 ⁽³⁷⁾ 05244397 ⁽³⁸⁾

Cable clips

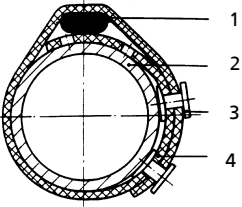
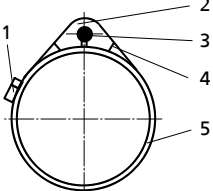
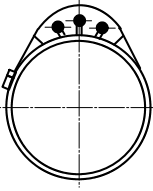
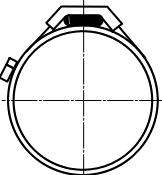
Application

- Securing electric cables at the riser / discharge pipe
- Suitable for use in normal or slightly aggressive water
- Versions for use in more aggressive water (e.g. seawater) on request

Table 64: Selection table

Electric cable			Cable clip size	Cable guard	Mat. No.	Weight	Required strap length L [mm] per cable clip for the following column pipes														
Type	Number x Ø of cores						[kg]	G 11/4 / DN 32	G 11/2 / DN 40	G 2 / DN 50	DN 65	G 3 / DN 80	G 4 / DN 100	G 5 / DN 125	G 6 / DN 150	DN 175	G 8 / DN 200	G 10 / DN 250	G 12 / DN 300	G 14 / DN 350	G 16 / DN 400
	[mm]																				
Flat	3 x 1.5 to 3 x 6.0	4 x 1.5 to 4 x 6.0	1	-	01088095	0.04	270	280	320	360	400	450	500	600	-	-	-	-	-	-	
	3 x 10 to 3 x 70	4 x 10 to 4 x 70	11	✓	On request	0,01	-	-	350	400	450	550	650	800	950	1100	1300	1500	1700	1900	
Round	3 x 1.5 to 3 x 6.0	4 x 1.5 to 4 x 6.0	1	-	01088095	0.04	270	280	320	360	400	450	500	600	-	-	-	-	-	-	
	3 x 10	4 x 10	2	✓	On request	0,08	-	-	350	400	450	550	650	800	950	1100	1300	1500	1700	1900	
	3 x 16 3 x 25	4 x 16	3a	-	On request	0,19	-	-	350	400	450	550	650	800	950	1100	1300	1500	1700	1900	
	3 x 35 3 x 50	4 x 25 4 x 35	3	-	On request	0,19	-	-	-	450	500	600	700	850	1000	1150	1350	1550	1750	1950	
	3 x 70	4 x 50	3b	-	On request	0,19	-	-	-	450	500	600	700	850	1000	1150	1350	1550	1750	1950	
	3 x 95	4 x 70 4 x 95	4	-	On request	0,55	-	-	-	-	-	650	750	900	1050	1200	1400	1600	1800	2000	
	3 cables 1 x 50		7	-	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050	
	3 cables 1 x 70		8	-	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050	
	3 cables 1 x 95		6	-	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050	
	3 cables 1 x 120		9	✓	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050	
	3 cables 1 x 150		9	✓	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050	

Table 65: Cable clip sizes

Size 1	Sizes 2, 3, 3a, 3b, 4	Sizes 6 to 9	Size 11
			
1 = electric cable 2 = riser 3 = plastic stud 4 = rubber strap	1 = screw band with tension screw (re-usable) 2 = cable guard 3 = electric cable 4 = metal strap 5 = riser		

Motor refill kit

Table 66: Overview of motor refill kit

	Description	Mat. No.	Weight
			[kg]
	Fill check kit for checking and topping up the motor fill after prolonged storage for UMA 150, UMA-S 150, UMA 200, UMA-S 200, UMA 250, UMA-S 250 motors	90066762	0.25
	Refill kit for checking and topping up the motor fill after prolonged storage for UMA 300 motor	Pipe (a)	0.01
		Filling funnel (b)	0.05

Monitoring and automation

Table 67: Selection table

Description		Mat. No.	Weight [kg/m]
Dry running protection device (supply voltage 220 V / 240 V) for			
	Semi-automatic operation (1 relay, 1 pushbutton + 2 electrodes)	90009553	2
	Fully automatic operation (1 relay + 3 electrodes)	90009554	2
	Plus 1 control cable, blue, 1 x 1.5 mm ² for each electrode; values indicated per metre	01096713	0.1
Immersion electrodes			
	Set of electrodes: 3 electrodes made of stainless steel	40980055	0.3
	Set of electrodes: 1 electrode made of stainless steel	40980056	0.1
Pressure controller , insulating enclosure (membrane type), 1 to 8 bar, with pressure connection G 3/8		01151586	1
Pt100 temperature monitoring equipment for use in submersible motors to protect the windings from overheating. Comprising:			
Pt100 temperature sensor with 10 m cable 4 x 0.5 mm ² for motor:	UMA 150 material variants E, C	01532593	1.1
	UMA 150 material variant D	01532594	1.1
	UMA 200, UMA 250, material variants G, C	90063006	1.1
	UMA 200, UMA 250, material variants D	On request	-
Display / switchgear (supply voltage V = 110/220 V)		90064446	0.45
Control cable, blue, round 4 x 0.5 mm ²		Per metre	01049403
			0.28

Frequency inverter and voltage filter

Table 68: Selection table

Motor UMA-S								Frequency inverter PumpDrive R				dv/dt filter				Sine filter			
Material variant				Size	P _N		I _N	Enclosure	Mat. No.	Designation	I _A	Enclosure	Mat. No.	Designation	I _{NFilter}	Enclosure	Mat. No.	Designation	I _{NFilter}
G	E	C	D		[kW]	[Hp]													
Mat. No.	Mat. No.	Mat. No.	Mat. No.		[kW]	[Hp]	[A]				[A]								
-	01639317	01732881	01732882	UMA-S 150 7/42	4,5	6	8	IP20	48229690	PumpDrive R KSB202 004K00C	10	IP00	01664325	130B2835	33	IP00	01664341	130B2411	13
-	01639317	01732881	01732882	UMA-S 150 7/42	5,5	7	10	IP20	48229692	PumpDrive R KSB202 005K50C	13	IP00	01664325	130B2835	33	IP00	01664341	130B2411	13
-	01639317	01732881	01732882	UMA-S 150 7/42	7,5	10	13	IP20	48229694	PumpDrive R KSB202 007K50C	16	IP00	01664325	130B2835	33	IP00	01664342	130B2412	18
-	01584165	01732855	01732856	UMA-S 150 18/42	9,3	12	18	IP20	48229696	PumpDrive R KSB202 011K00C	24	IP00	01664325	130B2835	33	IP00	01664343	130B2413	28,5
-	01584165	01732855	01732856	UMA-S 150 18/42	11	15	20	IP20	48229696	PumpDrive R KSB202 011K00C	24	IP00	01664325	130B2835	33	IP00	01664343	130B2413	28,5
-	01584165	01732855	01732856	UMA-S 150 18/42	13	17	23	IP20	48229698	PumpDrive R KSB202 015K00C	32	IP00	01664325	130B2835	33	IP00	01664344	130B2281	36
-	01584165	01732855	01732856	UMA-S 150 18/42	15	20	26	IP20	48229698	PumpDrive R KSB202 015K00C	32	IP00	01664325	130B2835	33	IP00	01664344	130B2281	36
-	01584165	01732855	01732856	UMA-S 150 18/42	18,5	25	32	IP20	48229700	PumpDrive R KSB202 018K50C	37,5	IP00	01664326	130B2838	68	IP00	01664345	130B2282	46,5
-	01584166	01732877	01732878	UMA-S 150 37/42	22	29	39	IP20	48229702	PumpDrive R KSB202 022K00C	44	IP00	01664326	130B2838	68	IP00	01664345	130B2282	46,5
-	01584166	01732877	01732878	UMA-S 150 37/42	26	35	46	IP20	48229704	PumpDrive R KSB202 030K00C	61	IP00	01664326	130B2838	68	IP00	01664346	130B3179	86
-	01584166	01732877	01732878	UMA-S 150 37/42	30	40	54	IP20	48229704	PumpDrive R KSB202 030K00C	61	IP00	01664326	130B2838	68	IP00	01664346	130B3179	86
-	01584197	01732879	01732880	UMA-S 150 37/42	37	49	72	IP20	48229708	PumpDrive R KSB202 045K00C	90	IP00	01674691	130B2844	133	IP00	01664346	130B3179	86
-	01639317	01732881	01732882	UMA-S 150 7/42	4,5	6	8	IP55	48229691	PumpDrive R KSB202 004K00W	10	IP54	01664338	130B2837	34	IP54	01664347	192H3025	17
-	01639317	01732881	01732882	UMA-S 150 7/42	5,5	7	10	IP55	48229693	PumpDrive R KSB202 005K50W	13	IP54	01664338	130B2837	34	IP54	01664347	192H3025	17
-	01639317	01732881	01732882	UMA-S 150 7/42	7,5	10	13	IP55	48229695	PumpDrive R KSB202 007K50W	16	IP54	01664338	130B2837	34	IP54	01664347	192H3025	17
-	01584165	01732855	01732856	UMA-S 150 18/42	9,3	12	18	IP55	48229697	PumpDrive R KSB202 011K00W	24	IP54	01664338	130B2837	34	IP54	01664348	192H3027	24,2
-	01584165	01732855	01732856	UMA-S 150 18/42	11	15	20	IP55	48229697	PumpDrive R KSB202 011K00W	24	IP54	01664338	130B2837	34	IP54	01664348	192H3027	24,2
-	01584165	01732855	01732856	UMA-S 150 18/42	13	17	23	IP55	48229699	PumpDrive R KSB202 015K00W	32	IP54	01664338	130B2837	34	IP54	01664349	192H3028	32
-	01584165	01732855	01732856	UMA-S 150 18/42	15	20	26	IP55	48229699	PumpDrive R KSB202 015K00W	32	IP54	01664338	130B2837	34	IP54	01664349	192H3028	32
-	01584165	01732855	01732856	UMA-S 150 18/42	18,5	25	32	IP55	48229701	PumpDrive R KSB202 018K50W	37,5	IP54	01664339	130B2840	56	IP54	01664350	192H3010	37,5

Motor UMA-S								Frequency inverter PumpDrive R				dv/dt filter				Sine filter			
Material variant				Size	P _N		I _N	Enclosure	Mat. No.	Designation	I _A	Enclosure	Mat. No.	Designation	I _{NFilter}	Enclosure	Mat. No.	Designation	I _{NFilter}
G	E	C	D		[kW]	[Hp]	[A]				[A]				[A]				[A]
Mat. No.	Mat. No.	Mat. No.	Mat. No.																
-	01584166	01732877	01732878	UMA-S 150 37/42	22	29	39	IP55	48229703	PumpDrive R KSB202 022K00W	44	IP54	01664339	130B2840	56	IP54	01664351	192H3011	46,2
-	01584166	01732877	01732878	UMA-S 150 37/42	26	35	46	IP55	48229705	PumpDrive R KSB202 030K00W	61	IP54	01664340	130B2843	83	IP54	01664352	192H3035	61
-	01584166	01732877	01732878	UMA-S 150 37/42	30	40	54	IP55	48229705	PumpDrive R KSB202 030K00W	61	IP54	01664340	130B2843	83	IP54	01664352	192H3035	61
-	01584197	01732879	01732880	UMA-S 150 37/42	37	49	72	IP55	48229709	PumpDrive R KSB202 045K00W	90	IP54	01664340	130B2843	83	IP54	01674692	192H3037	115
01738152	-	01738156	01738170	UMA-S 200 75/42	45	60	74	IP20	48229708	PumpDrive R KSB202 045K00C	90	IP00	01674691	130B2844	133	IP00	01664346	130B3179	86
01738152	-	01738156	01738170	UMA-S 200 75/42	55	74	91	IP20	48229710	PumpDrive R KSB202 055K00C	106	IP00	01674691	130B2844	133	IP00	01733111	130B3182	135
01738152	-	01738156	01738170	UMA-S 200 75/42	67	90	112	IP20	48229712	PumpDrive R KSB202 075K00C	147	IP00	01674691	130B2844	133	IP00	01733111	130B3182	135
01738152	-	01738156	01738170	UMA-S 200 75/42	75	100	128	IP20	48229714	PumpDrive R KSB202 090K00C	177	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195
01738153	-	01738167	01738171	UMA-S 200 100/42	75	100	129	IP20	48229714	PumpDrive R KSB202 090K00C	177	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195
01738153	-	01738167	01738171	UMA-S 200 100/42	83	110	143	IP20	48229714	PumpDrive R KSB202 090K00C	177	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195
01738153	-	01738167	01738171	UMA-S 200 100/42	93	125	163	IP20	48229716	PumpDrive R KSB202 110K00C	212	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195
01738153	-	01738167	01738171	UMA-S 200 100/42	100	135	178	IP20	48229716	PumpDrive R KSB202 110K00C	212	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195
05148497	-	05148498	05148551	UMA-S 200 130/42	110	145	186	IP20	01733781	PumpDrive R KSB202 132K00C	260	IP00	01733056	130B2847	236	IP00	01733113	130B3186	308
05148497	-	05148498	05148551	UMA-S 200 130/42	130	175	224	IP20	01733783	PumpDrive R KSB202 160K00C	315	IP00	01733056	130B2847	236	IP00	01733113	130B3186	308
01738152	-	01738156	01738170	UMA-S 200 75/42	45	60	74	IP55	48229709	PumpDrive R KSB202 045K00W	90	IP54	01733108	130B2846	133	IP54	01674692	192H3037	115
01738152	-	01738156	01738170	UMA-S 200 75/42	55	74	91	IP55	48229711	PumpDrive R KSB202 055K00W	106	IP54	01733108	130B2846	133	IP54	01674692	192H3037	115
01738152	-	01738156	01738170	UMA-S 200 75/42	67	90	112	IP55	48229713	PumpDrive R KSB202 075K00W	147	IP54	01733108	130B2846	133	IP54	01674692	192H3037	115
01738152	-	01738156	01738170	UMA-S 200 75/42	75	100	128	IP55	48229715	PumpDrive R KSB202 090K00W	177	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195
01738153	-	01738167	01738171	UMA-S 200 100/42	75	100	129	IP55	48229715	PumpDrive R KSB202 090K00W	177	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195
01738153	-	01738167	01738171	UMA-S 200 100/42	83	110	143	IP55	48229715	PumpDrive R KSB202 090K00W	177	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195
01738153	-	01738167	01738171	UMA-S 200 100/42	93	125	163	IP55	48229717	PumpDrive R KSB202 110K00W	212	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195

Motor UMA-S								Frequency inverter PumpDrive R				dv/dt filter				Sine filter			
Material variant				Size	P _N		I _N	Enclosure	Mat. No.	Designation	I _A	Enclosure	Mat. No.	Designation	I _{NFilter}	Enclosure	Mat. No.	Designation	I _{NFilter}
G	E	C	D		[kW]	[Hp]	[A]				[A]				[A]				[A]
Mat. No.	Mat. No.	Mat. No.	Mat. No.																
01738153	-	01738167	01738171	UMA-S 200 100/42	100	135	178	IP55	48229717	PumpDrive R KSB202 110K00W	212	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195
05148497	-	05148498	05148551	UMA-S 200 130/42	110	145	186	IP54	01733782	PumpDrive R KSB202 132K00W	260	IP23	01733109	130B2848	236	IP23	01733116	130B3187	308
05148497	-	05148498	05148551	UMA-S 200 130/42	130	175	224	IP54	01733784	PumpDrive R KSB202 160K00W	315	IP23	01733109	130B2848	236	IP23	01733116	130B3187	308
05147847	-	05147848	05147849	UMA-S 250 200/42	185	246	358	IP20	01839750	PumpDrive R KSB202 200K00C	395	IP00	01733107	130B2849	360	IP00	01845293	130B3188	382
05147847	-	05147848	05147849	UMA-S 250 200/42	200	266	389	IP20	01839789	PumpDrive R KSB202 250K00C	480	IP00	05146486	130B2851	494	IP00	05146604	130B3191	496
05147882	-	05147883	05147901	UMA-S 250 250/42	230	306	447	IP20	01839789	PumpDrive R KSB202 250K00C	480	IP00	05146486	130B2851	494	IP00	05146604	130B3191	496
05147882	-	05147883	05147901	UMA-S 250 250/42	250	333	493	IP20	05117684	PumpDrive R KSB202 315K00C	600	IP00	05146488	130B2853	660	IP00	05146622	130B3193	601
05147847	-	05147848	05147849	UMA-S 250 200/42	185	246	358	IP54	01839790	PumpDrive R KSB202 200K00W	395	IP23	01733110	130B2850	360	IP23	01845295	130B3189	382
05147847	-	05147848	05147849	UMA-S 250 200/42	200	266	389	IP54	01839792	PumpDrive R KSB202 250K00C	480	IP23	05146487	130B2852	494	IP23	05146605	130B3192	496
05147882	-	05147883	05147901	UMA-S 250 250/42	230	306	447	IP54	01839792	PumpDrive R KSB202 250K00C	480	IP23	05146487	130B2852	494	IP23	05146605	130B3192	496
05147882	-	05147883	05147901	UMA-S 250 250/42	250	333	493	IP54	05117687	PumpDrive R KSB202 315K00W	600	IP23	05146597	130B2854	660	IP23	05146623	130B3194	601

